Liu, Chapter 1: Some topics in commutative algebra

Solutions to the exercises

Conventions. Rings are commutative with 1. Results of the text are cited by their numbers in Liu ("Prop. 1.5", "Thm. 2.16", ...). The numbering and statements of the exercises follow the book (Exercises 1.1–1.7, 2.1–2.19, 3.1–3.15); each exercise is restated before its solution. A few sub-items that the book displays as nested lists are written as running prose.

Honest notes. Ex. 2.9 is false as printed for $M = 0$ (zero is flat but its localizations are free of rank 0), so it is proved for $M \neq 0$. Ex. 3.6 is proved for a nonzero proper ideal I (for $I = A$ the conclusion is trivial). Ex. 1.6 uses a directed index set, and Ex. 1.4(b) is read as "finitely generated as an A-algebra". In Ex. 3.3 the Taylor map on C^∞ germs is not injective, and the solution says what is true. Some solutions of Section 1.3 depend on earlier ones (3.7 uses 3.3; 3.10 uses 3.8, 3.9; 3.12 uses 3.10).

1.1 Tensor products

Throughout, A is a commutative ring with unit, all tensor products without subscript are over A, and results of the text are cited by their numbers in Liu ("Prop. 1.5", "Cor. 1.13", ...).

Exercise 1.1

Problem. Let $\{M_i\}_{i\in I}$, $\{N_j\}_{j\in J}$ be two families of modules over a ring A. Show that

$$\Big(\bigoplus_{i\in I} M_i\Big) \otimes_A \Big(\bigoplus_{j\in J} N_j\Big) \simeq \bigoplus_{(i,j)\in I\times J} (M_i \otimes_A N_j).$$

Solution. Write $M = \bigoplus_i M_i$ and $N = \bigoplus_j N_j$. We chain canonical isomorphisms from Prop. 1.5:

$$\begin{aligned}
M \otimes N &\simeq \bigoplus_{i\in I} (M_i \otimes N) && \text{by 1.5(d)},\\
M_i \otimes N &\simeq N \otimes M_i && \text{by 1.5(b)},\\
N \otimes M_i &\simeq \bigoplus_{j\in J} (N_j \otimes M_i) && \text{by 1.5(d)},\\
N_j \otimes M_i &\simeq M_i \otimes N_j && \text{by 1.5(b)}.
\end{aligned}$$

Each isomorphism in the second to fourth lines is applied inside a direct sum, componentwise (a direct sum of isomorphisms is an isomorphism). Composing, we get an isomorphism

$$M \otimes N \xrightarrow{\sim} \bigoplus_{i\in I}\bigoplus_{j\in J}(M_i \otimes N_j) = \bigoplus_{(i,j)\in I\times J}(M_i \otimes N_j).$$

Following a pure tensor $x \otimes y$ with $x = (x_i)_i$, $y = (y_j)_j$ through the chain, it goes to the family whose (i,j)-component is $x_i \otimes y_j$. So the isomorphism is the canonical one, characterized by sending $\iota_i(x_i) \otimes \iota_j(y_j)$ to the element $x_i \otimes y_j$ placed in the (i,j)-summand. □

Exercise 1.2

Problem. Show that there exists a unique A-linear map

$$f : \mathrm{Hom}_A(M, M') \otimes_A \mathrm{Hom}_A(N, N') \to \mathrm{Hom}_A(M \otimes_A N, M' \otimes_A N')$$

such that $f(u \otimes v) = u \otimes v$.

Solution. On the right, $u \otimes v$ denotes the map of Def. 1.8, the unique A-linear map $M \otimes N \to M' \otimes N'$ with $x \otimes y \mapsto u(x) \otimes v(y)$.

Existence. Define $g : \mathrm{Hom}_A(M, M') \times \mathrm{Hom}_A(N, N') \to \mathrm{Hom}_A(M \otimes N, M' \otimes N')$ by $g(u, v) = u \otimes v$. We check that g is A-bilinear. Let $u, u' \in \mathrm{Hom}_A(M, M')$, $v \in \mathrm{Hom}_A(N, N')$, $a \in A$. For $x \in M$, $y \in N$,

$$((u + u') \otimes v)(x \otimes y) = (u(x) + u'(x)) \otimes v(y) = (u \otimes v)(x \otimes y) + (u' \otimes v)(x \otimes y),$$

$$((au) \otimes v)(x \otimes y) = a\, u(x) \otimes v(y) = a\, (u \otimes v)(x \otimes y).$$

Both sides of each identity are A-linear maps on $M \otimes N$, and the elements $x \otimes y$ generate $M \otimes N$ (Rem. 1.3). Hence $(u + u') \otimes v = u \otimes v + u' \otimes v$ and $(au) \otimes v = a(u \otimes v)$. Linearity in v is identical. By the universal property of the tensor product (Def. 1.1) there is a unique A-linear f with $f(u \otimes v) = g(u, v) = u \otimes v$ for all u, v.

Uniqueness. The elements $u \otimes v$ generate the source (Rem. 1.3), so an A-linear map is determined by its values on them. This is also part of the uniqueness in Def. 1.1. □

Exercise 1.3

Problem. Let M, N be A-modules, and $i : M' \to M$, $j : N' \to N$ submodules of M and N, respectively. Then there exists a canonical isomorphism

$$(M/M') \otimes_A (N/N') \simeq (M \otimes_A N)/(\mathrm{Im}\, i_N + \mathrm{Im}\, j_M).$$

Show that $(\mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/m\mathbb{Z}) = \mathbb{Z}/l\mathbb{Z}$, where $l = \gcd(m, n)$.

Solution. Here $i_N = i \otimes \mathrm{Id}_N : M' \otimes N \to M \otimes N$ and $j_M = j \otimes \mathrm{Id}_M : N' \otimes M \to N \otimes M \simeq M \otimes N$. Hence $\mathrm{Im}\, i_N$ is the submodule of $M \otimes N$ generated by the $x' \otimes y$ ($x' \in M'$, $y \in N$), and $\mathrm{Im}\, j_M$ is generated by the $x \otimes y'$ ($x \in M$, $y' \in N'$). Put $K = \mathrm{Im}\, i_N + \mathrm{Im}\, j_M$.

Step 1. Let $\pi : M \to M/M'$ be the projection and $\pi_N = \pi \otimes \mathrm{Id}_N : M \otimes N \to (M/M') \otimes N$. By Cor. 1.13 (applied to $M' \to M$ and the module N), π_N is surjective with kernel $\mathrm{Im}\, i_N$.

Step 2. Let $q : N \to N/N'$ be the projection. The sequence $N' \xrightarrow{j} N \xrightarrow{q} N/N' \to 0$ is exact, so by Prop. 1.12 (tensoring with M/M', and using commutativity)

$$(M/M') \otimes N' \xrightarrow{\mathrm{Id}\otimes j} (M/M') \otimes N \xrightarrow{\mathrm{Id}\otimes q} (M/M') \otimes (N/N') \to 0$$

is exact. Thus $\mathrm{Id}\otimes q$ is surjective with kernel $W = \mathrm{Im}(\mathrm{Id}\otimes j)$. This kernel is generated by the elements $\bar{x} \otimes y'$ with $x \in M$, $y' \in N'$, which are the images under π_N of the generators $x \otimes y'$ of $\mathrm{Im}\, j_M$. Hence $W = \pi_N(\mathrm{Im}\, j_M)$.

Step 3. Consider the surjection $\Theta = (\mathrm{Id}\otimes q) \circ \pi_N : M \otimes N \to (M/M') \otimes (N/N')$, so that $\Theta(x \otimes y) = \bar{x} \otimes \bar{y}$. Its kernel is $\pi_N^{-1}(W) = \pi_N^{-1}(\pi_N(\mathrm{Im}\, j_M)) = \mathrm{Im}\, j_M + \ker \pi_N = \mathrm{Im}\, j_M + \mathrm{Im}\, i_N = K$. Therefore Θ induces an isomorphism

$$(M \otimes N)/K \xrightarrow{\sim} (M/M') \otimes (N/N'), \qquad \overline{x \otimes y} \mapsto \bar{x} \otimes \bar{y},$$

which is canonical.

The case of $\mathbb{Z}$. Take $A = \mathbb{Z}$. By Cor. 1.13 (with $I = n\mathbb{Z}$ and $M = \mathbb{Z}/m\mathbb{Z}$, and Prop. 1.5(b)),

$$(\mathbb{Z}/n\mathbb{Z}) \otimes (\mathbb{Z}/m\mathbb{Z}) \simeq (\mathbb{Z}/m\mathbb{Z})\big/n(\mathbb{Z}/m\mathbb{Z}).$$

The subgroup $n(\mathbb{Z}/m\mathbb{Z})$ of $\mathbb{Z}/m\mathbb{Z}$ is $(n\mathbb{Z} + m\mathbb{Z})/m\mathbb{Z} = l\mathbb{Z}/m\mathbb{Z}$, because $n\mathbb{Z} + m\mathbb{Z} = l\mathbb{Z}$. So the quotient is $(\mathbb{Z}/m\mathbb{Z})/(l\mathbb{Z}/m\mathbb{Z}) \simeq \mathbb{Z}/l\mathbb{Z}$. (When $l = 1$ this recovers Example 1.4.) □

Exercise 1.4

Problem. Let M, N be A-modules, and let B, C be A-algebras.

(a) If M and N are finitely generated over A, then so is $M \otimes_A N$.

(b) If B and C are finitely generated over A, then so is $B \otimes_A C$.

(c) Taking $A = \mathbb{Z}$, $M = B = \mathbb{Z}/2\mathbb{Z}$, and $N = C = \mathbb{Q}$, show that the converse of (a) and (b) is false.

Solution. In (b), "finitely generated over A" means finitely generated as an A-algebra. (If B, C are finitely generated as A-modules, then so is $B \otimes_A C$ by (a); this is a stronger hypothesis and a stronger conclusion, which gives the module version of (b).)

(a) Let $x_1, \ldots, x_r$ generate M and $y_1, \ldots, y_s$ generate N. By Rem. 1.3, $M \otimes N$ is generated by the elements $x \otimes y$. Write $x = \sum_k a_k x_k$ and $y = \sum_l b_l y_l$. Then $x \otimes y = \sum_{k,l} a_k b_l\, (x_k \otimes y_l)$. So the rs elements $x_k \otimes y_l$ generate $M \otimes N$.

(b) Let $B = A[b_1, \ldots, b_r]$ and $C = A[c_1, \ldots, c_s]$ be generated as algebras. Let $D \subseteq B \otimes C$ be the A-subalgebra generated by the $b_k \otimes 1 = p_1(b_k)$ and $1 \otimes c_l = p_2(c_l)$. Since p_1, p_2 are A-algebra homomorphisms, D contains $p_1(B)$ and $p_2(C)$: for a polynomial P with coefficients in A, $p_1(P(b_1, \ldots, b_r)) = P(b_1 \otimes 1, \ldots, b_r \otimes 1) \in D$, and similarly for p_2. Hence for $b \in B$, $c \in C$ we get $b \otimes c = (b \otimes 1)(1 \otimes c) = p_1(b)p_2(c) \in D$. Since the elements $b \otimes c$ generate $B \otimes C$ additively (Rem. 1.3), $D = B \otimes C$. So $B \otimes_A C$ is generated by the $r + s$ elements $b_k \otimes 1$, $1 \otimes c_l$.

(c) Let $A = \mathbb{Z}$. For $x \in \mathbb{Z}/2\mathbb{Z}$ and $y \in \mathbb{Q}$ we have $x \otimes y = x \otimes 2(y/2) = (2x) \otimes (y/2) = 0$, since $2x = 0$. As such elements generate (Rem. 1.3), $M \otimes N = (\mathbb{Z}/2\mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Q} = 0$. The same computation gives $B \otimes_{\mathbb{Z}} C = 0$. The zero module is finitely generated over $\mathbb{Z}$, and the zero ring is generated as a $\mathbb{Z}$-algebra by $\emptyset$ (or by 0). Also $M = B = \mathbb{Z}/2\mathbb{Z}$ is finitely generated as a module (by 1) and as an algebra. But $\mathbb{Q}$ is neither:

- $\mathbb{Q}$ *is not a finitely generated* $\mathbb{Z}$*-module.* If $q_1, \ldots, q_r \in \mathbb{Q}$ generate $\mathbb{Q}$, choose a common denominator $d \geq 1$ with $dq_k \in \mathbb{Z}$ for all k. Then every element of $\mathbb{Q}$ lies in $\frac{1}{d}\mathbb{Z}$. But $\frac{1}{2d} \notin \frac{1}{d}\mathbb{Z}$, a contradiction.
- $\mathbb{Q}$ *is not a finitely generated* $\mathbb{Z}$*-algebra.* Suppose $\mathbb{Q} = \mathbb{Z}[a_1/b_1, \ldots, a_r/b_r]$ with $a_k, b_k \in \mathbb{Z}$, $b_k \neq 0$. Put $b = b_1 \cdots b_r$. Then $\mathbb{Q} \subseteq \mathbb{Z}[1/b] = \{m/b^t : m \in \mathbb{Z},\ t \geq 0\}$. Take a prime $p \nmid b$. If $1/p = m/b^t$, then $b^t = pm$, so $p \mid b^t$ and hence $p \mid b$, a contradiction. So $1/p \notin \mathbb{Z}[1/b]$.

Therefore $M \otimes N$ and $B \otimes C$ are finitely generated while N and C are not, so the converses of (a) and (b) fail. □

Exercise 1.5

Problem. Let $\rho : A \to B$ be a ring homomorphism, M an A-module, and N a B-module. Show that there exists a canonical isomorphism of A-modules

$$\mathrm{Hom}_A(M, \rho_* N) \simeq \mathrm{Hom}_B(\rho^* M, N).$$

Solution. Recall $\rho_* N$ is N with $a \cdot y = \rho(a)y$, and $\rho^* M = M \otimes_A B$ with B-action $b'(x \otimes b) = x \otimes b'b$. The right side is a B-module, hence an A-module through ρ; the left side is an A-module in the usual way.

The map Φ. Let $g \in \mathrm{Hom}_A(M, \rho_* N)$. The map $M \times B \to N$, $(x, b) \mapsto b\, g(x)$, is A-bilinear. It is biadditive. It is compatible with scalars: for $a \in A$, $(ax, b) \mapsto b\, g(ax) = b\rho(a)g(x) = (\rho(a)b)g(x)$ and $(x, \rho(a)b) \mapsto (\rho(a)b)g(x)$, and $(x, ab) \mapsto \rho(a)b\, g(x)$ also (here $a \cdot b = \rho(a)b$ in B). So it induces an A-linear map $\tilde{g} : M \otimes_A B \to N$ with $\tilde{g}(x \otimes b) = b\, g(x)$. It is B-linear: $\tilde{g}(b'(x \otimes b)) = \tilde{g}(x \otimes b'b) = b'b\, g(x) = b'\tilde{g}(x \otimes b)$, and the identity on generators suffices. Put $\Phi(g) = \tilde{g}$.

The map Ψ. For $h \in \mathrm{Hom}_B(\rho^* M, N)$ put $\Psi(h)(x) = h(x \otimes 1)$. It is A-linear into $\rho_* N$: additivity is clear and

$$h(ax \otimes 1) = h(x \otimes \rho(a)) = h(\rho(a)(x \otimes 1)) = \rho(a)h(x \otimes 1) = a \cdot h(x \otimes 1).$$

Inverse. $\Psi\Phi(g)(x) = \tilde{g}(x \otimes 1) = g(x)$. And $\Phi\Psi(h)(x \otimes b) = b\, h(x \otimes 1) = h(b(x \otimes 1)) = h(x \otimes b)$; since the $x \otimes b$ generate $M \otimes_A B$, $\Phi\Psi(h) = h$. So Φ is bijective with inverse Ψ.

A-linearity. Φ is clearly additive. For $a \in A$, $\Phi(ag)(x \otimes b) = b\, \rho(a)g(x) = \rho(a)\, \Phi(g)(x \otimes b) = (a\Phi(g))(x \otimes b)$. Hence Φ is an isomorphism of A-modules. It involves no choices, so it is canonical. □

Exercise 1.6

Problem. Let $(N_i)_{i\in I}$ be a direct system of A-modules. Then for any A-module M, there exists a canonical isomorphism $\varinjlim(N_i \otimes_A M) \simeq (\varinjlim N_i) \otimes_A M$.

Solution. As usual, a direct system is indexed by a *directed* set I (for $i, j \in I$ there is $k \geq i, j$), with transition maps $\varphi_{ik} : N_i \to N_k$ for $i \leq k$. (For a non-directed index set the statement is still true, but the argument below, which uses the elementwise description of the limit, needs directedness.) We use the standard facts about $N = \varinjlim N_i$ with canonical maps $\lambda_i : N_i \to N$: every element of N is of the form $\lambda_i(n_i)$; and $\lambda_i(n_i) = \lambda_j(n_j)$ if and only if $\varphi_{ik}(n_i) = \varphi_{jk}(n_j)$ for some $k \geq i, j$. Let $P = \varinjlim(N_i \otimes M)$, for the system with transition maps $\varphi_{ik} \otimes \mathrm{Id}_M$, and canonical maps $\mu_i : N_i \otimes M \to P$. Every element of P is of the form $\mu_i(z)$ with $z \in N_i \otimes M$.

The map $\alpha : P \to N \otimes M$. The maps $\lambda_i \otimes \mathrm{Id}_M : N_i \otimes M \to N \otimes M$ are compatible with the transition maps, because $\lambda_k \circ \varphi_{ik} = \lambda_i$. By the universal property of the direct limit they induce α with $\alpha\mu_i(n_i \otimes m) = \lambda_i(n_i) \otimes m$.

The map $\beta : N \otimes M \to P$. Define $b : N \times M \to P$ by $b(n, m) = \mu_i(n_i \otimes m)$ where $n = \lambda_i(n_i)$. This is well defined. If also $n = \lambda_j(n_j)$, pick $k \geq i, j$ with $\varphi_{ik}(n_i) = \varphi_{jk}(n_j)$. Then

$$\mu_i(n_i \otimes m) = \mu_k(\varphi_{ik}(n_i) \otimes m) = \mu_k(\varphi_{jk}(n_j) \otimes m) = \mu_j(n_j \otimes m),$$

using $\mu_k \circ (\varphi_{ik} \otimes \mathrm{Id}) = \mu_i$. It is bilinear. Given $n, n' \in N$, write them as $\lambda_i(n_i), \lambda_i(n_i')$ with a common index i (possible by directedness, after pushing forward to a common k). Then $n + n' = \lambda_i(n_i + n_i')$, so $b(n + n', m) = \mu_i((n_i + n_i') \otimes m) = b(n, m) + b(n', m)$. Additivity in m and compatibility with scalars $a \in A$ follow in the same way from $an = \lambda_i(an_i)$ and the bilinearity of $\otimes$. So b induces an A-linear β with $\beta(n \otimes m) = \mu_i(n_i \otimes m)$.

Inverse isomorphisms. For $n_i \in N_i$, $m \in M$ we have $\beta\alpha\mu_i(n_i \otimes m) = \beta(\lambda_i(n_i) \otimes m) = \mu_i(n_i \otimes m)$. The elements $\mu_i(n_i \otimes m)$ generate P (every element of P is some $\mu_i(z)$ with z a sum of pure tensors), so $\beta\alpha = \mathrm{Id}_P$. Also $\alpha\beta(n \otimes m) = \alpha\mu_i(n_i \otimes m) = \lambda_i(n_i) \otimes m = n \otimes m$, and the $n \otimes m$ generate $N \otimes M$, so $\alpha\beta = \mathrm{Id}$. Hence α is an isomorphism, and it is canonical (it sends $\mu_i(n_i \otimes m)$ to $\lambda_i(n_i) \otimes m$).

Remark on the hint. Equivalently, P together with the bilinear map b satisfies the universal property of $N \otimes M$, and the computation above is exactly the verification. □

Exercise 1.7

Problem. Let B be an A-algebra, and let M, N be B-modules. Show that there exists a canonical surjective homomorphism $M \otimes_A N \to M \otimes_B N$.

Solution. Let $\rho : A \to B$ be the structure map; M, N are A-modules via ρ, so $ax = \rho(a)x$. Consider the map $\theta : M \times N \to M \otimes_B N$, $\theta(x, y) = x \otimes_B y$. It is biadditive, and for $a \in A$,

$$\theta(ax, y) = \rho(a)x \otimes_B y = \rho(a)(x \otimes_B y) = x \otimes_B \rho(a)y = \theta(x, ay) = a\,\theta(x, y).$$

So θ is A-bilinear, and by Def. 1.1 there is a unique A-linear map $\Theta : M \otimes_A N \to M \otimes_B N$ with $\Theta(x \otimes_A y) = x \otimes_B y$. It is canonical, as it is characterized by this formula (it is even B-linear). The B-module structure on $M \otimes_A N$ is given by $b(x \otimes_A y) = bx \otimes_A y$, and Θ is B-linear: $\Theta(b(x \otimes_A y)) = bx \otimes_B y = b\,\Theta(x \otimes_A y)$. Hence the image of Θ is a B-submodule of $M \otimes_B N$ containing every $x \otimes_B y$. A B-submodule containing a B-generating set is everything, so Θ is surjective. □

1.2 Flatness

Throughout, rings are commutative with 1. We use the following elementary fact, which is not stated in the text.

(D) *A direct summand of a flat module is flat.* Indeed, let $M = M_1 \oplus M_2$ be flat and let $u : N' \to N$ be injective. By Prop. 1.5 the map $u \otimes \mathrm{id}_M : N' \otimes M \to N \otimes M$ is identified with the direct sum of the maps $N' \otimes M_i \to N \otimes M_i$; a direct sum of maps is injective only if each summand is injective, so M_1 and M_2 are flat.

Exercise 2.1

> **Problem.** Let M be an A-module, $\mathrm{Ann}(M) = \{a \in A \mid aM = 0\}$ its annihilator, and $I \subseteq \mathrm{Ann}(M)$ an ideal.
>
> (a) Show that M is endowed, in a natural way, with the structure of an A/I-module, and that $M \simeq M \otimes_A A/I$.
>
> (b) Let N be another A-module such that $I \subseteq \mathrm{Ann}(N)$. Show that the canonical homomorphism $M \otimes_A N \to M \otimes_{A/I} N$ is an isomorphism.

Solution. (a) Define $(a + I) \cdot m = am$. This is well defined: if $a - a' \in I$ then $(a - a')m = 0$. The module axioms for A/I follow from those for A. By Cor. 1.13, $M \otimes_A A/I \simeq M/IM$, and $IM = 0$ because $I \subseteq \mathrm{Ann}(M)$. So $M \otimes_A A/I \simeq M$ (explicitly, $m \otimes \bar{a} \mapsto am$).

(b) By Exercise 1.7 there is a canonical surjection $\alpha : M \otimes_A N \to M \otimes_{A/I} N$, $m \otimes n \mapsto m \otimes n$. Conversely, the map $M \times N \to M \otimes_A N$, $(m, n) \mapsto m \otimes n$, is A-bilinear, and it is A/I-bilinear for the structures of (a): $(\bar{a}m) \otimes n = am \otimes n = m \otimes an = m \otimes \bar{a}n$. Hence it induces an additive map $\beta : M \otimes_{A/I} N \to M \otimes_A N$ with $\beta(m \otimes n) = m \otimes n$. Now $\alpha\beta$ and $\beta\alpha$ are the identity on the generators $m \otimes n$ of the respective tensor products, hence are the identity. So α is an isomorphism. □

Exercise 2.2

> **Problem.** Let $\rho : A \to B$ be a ring homomorphism, S a multiplicative subset of A, and $T = \rho(S)$. Show that T is a multiplicative subset of B, and that $T^{-1}B \simeq B \otimes_A S^{-1}A$ as A-algebras.

Solution. Since $\rho(1) = 1$ and $\rho(s)\rho(s') = \rho(ss')$, T is multiplicative. Regard B as an A-module. By Lemma 2.10 there is an isomorphism of $S^{-1}A$-modules $B \otimes_A S^{-1}A \simeq S^{-1}B$, $b \otimes a/s \mapsto \rho(a)b/s$, where $S^{-1}B$ is the localization of the A-module B: elements b/s with $b/s = b'/s'$ iff $s''(s'b - sb') = 0$ for some $s'' \in S$, i.e. $\rho(s'')(\rho(s')b - \rho(s)b') = 0$ in B. This is exactly the equivalence relation defining $T^{-1}B$ for $b/\rho(s)$ and $b'/\rho(s')$, with $\rho(s'') \in T$ (every element of T is of the form $\rho(s'')$). Hence $b/s \mapsto b/\rho(s)$ is a well-defined bijection $S^{-1}B \to T^{-1}B$, compatible with addition (both have $b/s + b'/s' = (s'b + sb')/(ss')$). Composing, we get a bijection

$$\theta : B \otimes_A S^{-1}A \to T^{-1}B, \qquad b \otimes \tfrac{a}{s} \mapsto \tfrac{\rho(a)b}{\rho(s)}.$$

It is a ring homomorphism: the ring structure on $B \otimes_A S^{-1}A$ is given by $(b \otimes x)(b' \otimes x') = bb' \otimes xx'$, and θ is multiplicative on such pure tensors because $\frac{\rho(a)b}{\rho(s)} \cdot \frac{\rho(a')b'}{\rho(s')} = \frac{\rho(aa')bb'}{\rho(ss')}$; $\theta(1) = 1$. It is compatible with the maps from A: $a \mapsto 1 \otimes a/1 \mapsto \rho(a)/1$, and $a \mapsto \rho(a) \otimes 1$ also maps to $\rho(a)/1$ (the two A-algebra structures on $B \otimes_A S^{-1}A$ through B and through A coincide, as $\rho(a) \otimes 1 = a \cdot (1 \otimes 1) = 1 \otimes a/1$). So θ is an isomorphism of A-algebras (indeed of B-algebras and of $S^{-1}A$-algebras). □

Exercise 2.3

Problem. Show that Nakayama's lemma is false for modules M that are not finitely generated.

Solution. Let $A = \mathbf{Z}_{(p)} = \{a/b \in \mathbf{Q} \mid p \nmid b\}$, a local ring with maximal ideal $\mathfrak{m} = pA = \mathrm{Rad}(A)$, and let $M = \mathbf{Q}$, regarded as an A-module. Then $\mathfrak{m}M = pM = M$, since $x = p \cdot (x/p)$ for all $x \in \mathbf{Q}$; but $M \neq 0$. So the statement of Lemma 2.7 (with $I = \mathrm{Rad}(A)$) fails for M. (Note that M is not finitely generated, as any finitely generated A-submodule of $\mathbf{Q}$ is of the form $p^{-k}A$.) □

Exercise 2.4

Problem. Let I be a finitely generated ideal of A. Show that the following properties are equivalent: (i) A/I is flat over A; (ii) $I = I^2$; (iii) there exists $e \in A$ such that $e^2 = e$ and $I = eA$.

Solution. (i)⇒(ii). Let $M = A/I$ be flat. By Thm. 2.4 the canonical map $I \otimes_A M \to IM$ is an isomorphism. Now $IM = I(A/I) = 0$, while $I \otimes_A A/I \simeq I/I^2$ by Cor. 1.13. Hence $I/I^2 = 0$, i.e. $I = I^2$.

(ii)⇒(iii). Let $I = (x_1, \ldots, x_n)$. Since $I = I^2$, we can write $x_i = \sum_j a_{ij}x_j$ with $a_{ij} \in I$. Let $\Delta = \det(\delta_{ij} - a_{ij})$. Multiplying the system $\sum_j (\delta_{ij} - a_{ij})x_j = 0$ by the adjugate matrix gives $\Delta x_j = 0$ for all j, so $\Delta I = 0$. Expanding the determinant, $\Delta = 1 - e$ with $e \in I$. Then for every $x \in I$, $(1-e)x = 0$, i.e. $x = ex \in eA$. Thus $I \subseteq eA$, and $eA \subseteq I$ since $e \in I$. Finally $e \in I$ gives $e = ee$, so $e^2 = e$.

(iii)⇒(i). Let $e^2 = e$, $I = eA$. Then $A = eA \oplus (1-e)A$ (any $a = ea + (1-e)a$, and if $ea = (1-e)b$ then multiplying by e gives $ea = 0$). The composite $(1-e)A \to A \to A/I$ is surjective (since $a \equiv (1-e)a \bmod I$) and injective (if $(1-e)a = eb$, multiplying by $1-e$ gives $(1-e)a = 0$). So $A/I \simeq (1-e)A$ is a direct summand of the free, hence flat (Prop. 2.2(a)), module A, and is flat by (D). □

Exercise 2.5

Problem. Let A be an integral domain. Show that every A-module is flat if and only if A is a field.

Solution. If A is a field, every A-module is a vector space, hence free, hence flat (Prop. 2.2(a)).

Conversely, suppose every A-module is flat. Let $a \in A$ be nonzero and $I = aA$. Then A/I is flat and I is finitely generated, so by Exercise 2.4, $I = I^2$. Thus $a = a^2 b$ for some $b \in A$, i.e. $a(1 - ab) = 0$. As A is a domain and $a \neq 0$, $ab = 1$. So every nonzero element is a unit, and A (which is a domain, so $1 \neq 0$) is a field. $\square$

Exercise 2.6

Problem. Let B be a flat A-algebra, with structure map $\rho : A \to B$.

(a) Show that for any finite family $\{I_\lambda\}_{\lambda \in \Lambda}$ of ideals of A, we have $\bigcap_{\lambda \in \Lambda}(I_\lambda B) = \big(\bigcap_{\lambda \in \Lambda} I_\lambda\big)B$.

(b) Suppose B is faithfully flat. Show that $IB \cap A = I$ for any ideal I of A (here $IB \cap A$ means $\rho^{-1}(IB)$).

(c) Let k be a field, $A = k[t, s]$ and $C = A[z]/(tz - s)$. By considering the ideals tA and sA, show that C is not flat over A.

Solution. (a) The inclusion $\supseteq$ is clear. For $\subseteq$ we may assume $\Lambda \neq \emptyset$ (for $\Lambda = \emptyset$ both sides are B) and, by induction on $|\Lambda|$, it suffices to treat two ideals I, J: indeed $\bigcap_\lambda I_\lambda B = I_1 B \cap \big(\bigcap_{\lambda \neq 1} I_\lambda B\big) = I_1 B \cap \big(\bigcap_{\lambda \neq 1} I_\lambda\big)B = (I_1 \cap \bigcap_{\lambda \neq 1} I_\lambda)B$ by the case of two ideals.

For two ideals, consider the exact sequence of A-modules

$$0 \to A/(I \cap J) \xrightarrow{x \mapsto (x,x)} A/I \oplus A/J.$$

Tensoring with the flat A-module B and using $A/\mathfrak{a} \otimes_A B \simeq B/\mathfrak{a}B$ (Cor. 1.13) and Prop. 1.5, we get an exact sequence

$$0 \to B/(I \cap J)B \to B/IB \oplus B/JB, \qquad \bar{b} \mapsto (\bar{b}, \bar{b}).$$

If $b \in IB \cap JB$, then $\bar{b}$ maps to 0, so $b \in (I \cap J)B$. Thus $IB \cap JB \subseteq (I \cap J)B$.

(b) Let $J = \rho^{-1}(IB)$, an ideal of A with $I \subseteq J$ and $\rho(J) \subseteq IB$, hence $JB \subseteq IB$; also $IB \subseteq JB$. So $IB = JB$. Let $f : A/I \to A/J$ be the canonical surjection. Under the identifications $A/\mathfrak{a} \otimes_A B = B/\mathfrak{a}B$ (Cor. 1.13), $f \otimes \mathrm{id}_B$ is the canonical map $B/IB \to B/JB$, which is an isomorphism because $IB = JB$. Since B is faithfully flat, Prop. 2.17(iii) shows that f is an isomorphism, i.e. $J = I$. So $IB \cap A = I$.

(c) Since $s = tz$ in C, we have $C = k[t, s, z]/(tz - s) \simeq k[t, z]$ with $t \mapsto t$, $s \mapsto tz$ (eliminate the variable s). Suppose C is flat over A. By (a), $(tA \cap sA)C = tC \cap sC$. Now $tA \cap sA = tsA$ (A is a UFD and t, s are non-associate primes), so the left side is $t \cdot tz\,C = t^2 zC$. The right side is $tC \cap tzC = tzC$ (as $tzC \subseteq tC$). But $tz \notin t^2 zC$: otherwise $tz = t^2 zg$ for some $g \in k[t, z]$, so $1 = tg$ in the domain $k[t, z]$, impossible. This contradiction shows that C is not flat over A. $\square$

Exercise 2.7

Problem. Give an example of a finitely generated flat module that is not free (over a suitable ring A).

Solution. Let k be a field, $A = k \times k$, $e = (1, 0)$, and $M = eA = k \times 0$. Then $e^2 = e$, $A = eA \oplus (1 - e)A$ (as in Exercise 2.4), so M is a direct summand of the free module A and hence flat by Prop. 2.2(a) and (D). It is finitely generated (by e). It is not free: it is nonzero, so a free structure would have rank ≥ 1, and then M would contain a copy of A, forcing $\mathrm{Ann}(M) = 0$. But $(0, 1) \cdot M = 0$ with $(0, 1) \neq 0$. Hence M is flat, finitely generated, and not free. (Note that A is not local, in accordance with Thm. 2.16.) $\square$

Exercise 2.8

Problem. Let A be a Noetherian ring, M a finitely generated A-module, N an A-module, and B a flat A-algebra. Consider the canonical homomorphism

$$\rho : \mathrm{Hom}_A(M, N) \otimes_A B \to \mathrm{Hom}_B(M \otimes_A B, N \otimes_A B).$$

(a) Show that ρ is an isomorphism if M is free of finite rank.

(b) Let $0 \to K \to L \to M \to 0$ be an exact sequence of A-modules. Show that we have a canonical exact sequence $0 \to \mathrm{Hom}_A(M, N) \to \mathrm{Hom}_A(L, N) \to \mathrm{Hom}_A(K, N)$.

(c) By taking for L a free module of finite rank, show that ρ is injective. By applying the injectivity to K, show that ρ is an isomorphism.

Solution. Here $\rho(\varphi \otimes b)$ is the B-linear map $m \otimes b' \mapsto \varphi(m) \otimes bb'$. We write ρ_M to indicate the dependence on M (N, B being fixed); ρ_M is defined for every A-module M, and it is natural in M: if $u : M_1 \to M_2$ is A-linear, the square formed by ρ_{M_2}, ρ_{M_1} and the maps $u^* \otimes \mathrm{id}_B$, $(u \otimes \mathrm{id}_B)^*$ commutes, as one checks on the generators $\varphi \otimes b$ of $\mathrm{Hom}_A(M_2, N) \otimes B$: both ways give $m \otimes b' \mapsto \varphi(u(m)) \otimes bb'$.

(a) Let $M = A^n$ with basis $e_1, \ldots, e_n$. Then $\mathrm{Hom}_A(M, N) \simeq N^n$ via $\varphi \mapsto (\varphi(e_i))_i$; $M \otimes B \simeq B^n$ has the B-basis $e_i \otimes 1$, so $\mathrm{Hom}_B(M \otimes B, N \otimes B) \simeq (N \otimes B)^n$. Under these identifications ρ becomes $N^n \otimes_A B \to (N \otimes_A B)^n$, $(n_i)_i \otimes b \mapsto (n_i \otimes b)_i$, which is the canonical isomorphism of Prop. 1.5 (tensor product commutes with finite direct sums). Hence ρ is an isomorphism.

(b) Let $j : K \to L$ and $p : L \to M$. The map $p^* : \mathrm{Hom}(M, N) \to \mathrm{Hom}(L, N)$, $\psi \mapsto \psi p$, is injective because p is surjective. Next $j^* p^* \psi = \psi p j = 0$. Conversely, if $\varphi \in \mathrm{Hom}(L, N)$ satisfies $\varphi j = 0$, then φ vanishes on $K = \mathrm{Ker}\, p$ and so factors as $\varphi = \psi p$ with $\psi \in \mathrm{Hom}(M, N)$ (since p induces $L/K \simeq M$). So $\mathrm{Ker}\, j^* = \mathrm{Im}\, p^*$.

(c) Since M is finitely generated, there is an exact sequence $0 \to K \to L \to M \to 0$ with $L = A^n$ free of finite rank; K is finitely generated since A is Noetherian. By (b) and flatness of B (which preserves exactness) we get an exact row

$$0 \to \mathrm{Hom}_A(M, N) \otimes B \to \mathrm{Hom}_A(L, N) \otimes B \to \mathrm{Hom}_A(K, N) \otimes B.$$

Also $0 \to K \otimes B \to L \otimes B \to M \otimes B \to 0$ is exact because B is flat, so (b) over B applied to the B-module $N \otimes B$ gives an exact row

$$0 \to \mathrm{Hom}_B(M_B, N_B) \to \mathrm{Hom}_B(L_B, N_B) \to \mathrm{Hom}_B(K_B, N_B),$$

where $X_B = X \otimes_A B$. The maps ρ_M, ρ_L, ρ_K form a commutative diagram between these rows (naturality), and ρ_L is an isomorphism by (a).

Injectivity of ρ_M. Let $x \in \mathrm{Hom}_A(M,N) \otimes B$ with $\rho_M(x) = 0$ and let x_L be its image in $\mathrm{Hom}_A(L,N) \otimes B$. Then $\rho_L(x_L)$ is the image of $\rho_M(x) = 0$, so $x_L = 0$ as ρ_L is injective; by exactness of the first row (injectivity of the first arrow), $x = 0$. This argument applies to every finitely generated A-module M (with such a presentation); in particular to K, so ρ_K is injective.

Surjectivity of ρ_M. Let $y \in \mathrm{Hom}_B(M_B, N_B)$ and let y_L be its image in $\mathrm{Hom}_B(L_B, N_B)$. Write $y_L = \rho_L(z)$ with $z \in \mathrm{Hom}_A(L,N) \otimes B$. The image z_K of z in $\mathrm{Hom}_A(K,N) \otimes B$ satisfies $\rho_K(z_K) =$ image of y_L in $\mathrm{Hom}_B(K_B, N_B)$, which is 0 by exactness of the second row. As ρ_K is injective, $z_K = 0$, so by exactness of the first row z is the image of some $x \in \mathrm{Hom}_A(M,N) \otimes B$. Then $\rho_M(x)$ and y have the same image $\rho_L(z) = y_L$ in $\mathrm{Hom}_B(L_B, N_B)$, and the map $\mathrm{Hom}_B(M_B, N_B) \to \mathrm{Hom}_B(L_B, N_B)$ is injective, so $\rho_M(x) = y$. Thus ρ is an isomorphism. □

Exercise 2.9

> **Problem.** Let A be an integral domain, K its field of fractions, and M a finitely generated sub-A-module of K. Show that M is flat if and only if it is locally free of rank 1 (i.e. $M_\mathfrak{p}$ is free of rank 1 over $A_\mathfrak{p}$ for every prime ideal $\mathfrak{p}$ of A).

Solution. *Correction.* For $M = 0$ the statement is false as printed: 0 is flat, but $M_\mathfrak{p} = 0$ is free of rank 0, not 1. We prove the statement for $M \neq 0$ (for $M = 0$ the correct statement is that $M_\mathfrak{p}$ is free of rank 0 for all $\mathfrak{p}$).

Since K is torsion-free over A, the localization $M_\mathfrak{p}$ is identified with the $A_\mathfrak{p}$-submodule $A_\mathfrak{p}M = \{x/s\}$ of K: the relation $s''(s'm - sm') = 0$ in $M \subseteq K$ with $s'' \neq 0$ is equivalent to $s'm = sm'$.

($\Leftarrow$) If $M_\mathfrak{p}$ is free over $A_\mathfrak{p}$ for every prime $\mathfrak{p}$, in particular for every maximal ideal $\mathfrak{m}$, then $M_\mathfrak{m}$ is flat over $A_\mathfrak{m}$ (Prop. 2.2(a)), so M is flat over A by Prop. 2.13 ((iii)$\Rightarrow$(i)).

($\Rightarrow$) Let $M \neq 0$ be flat and $\mathfrak{p}$ a prime. Then $M_\mathfrak{p}$ is flat over $A_\mathfrak{p}$ (Prop. 2.13) and finitely generated, and $A_\mathfrak{p}$ is local, so $M_\mathfrak{p}$ is free by Thm. 2.16. Since $M \neq 0$, pick $0 \neq x \in M$; then $x/1 \in M_\mathfrak{p} \subseteq K$ is nonzero, so $M_\mathfrak{p} \neq 0$ and its rank is ≥ 1. If the rank were ≥ 2, there would be two elements u, v of a basis of $M_\mathfrak{p}$, nonzero elements of K, say $u = a/b$ and $v = c/d$ with $a, b, c, d \in A \setminus \{0\}$. Then $(cb)u - (ad)v = ca - ac = 0$ with $cb \neq 0$, a nontrivial linear relation over $A_\mathfrak{p}$ (as $A_\mathfrak{p} \subseteq K$ is a domain), contradicting linear independence. So $M_\mathfrak{p}$ is free of rank 1. □

Exercise 2.10

> **Problem.** Let A be an integral domain and B its integral closure in the field of fractions $\mathrm{Frac}(A)$. Suppose B is a finitely generated A-module. Show that B is flat over A if and only if $B = A$. (One can show that this is true without the finiteness assumption.)

Solution. If $B = A$ then B is free of rank 1 over A, hence flat.

Conversely let B be flat. Put $K = \mathrm{Frac}(A)$; then $A \subseteq B \subseteq K$, and $B \neq 0$ is a finitely generated A-submodule of K. By Exercise 2.9, for every prime $\mathfrak{p}$ of A the $A_\mathfrak{p}$-module $B_\mathfrak{p} = A_\mathfrak{p}B \subseteq K$ is free of rank 1: $B_\mathfrak{p} = xA_\mathfrak{p}$ for some $x \in K^\times$. Since $1 \in A \subseteq B_\mathfrak{p}$, we can write $1 = ax$ with $a \in A_\mathfrak{p}$. Every element of $B_\mathfrak{p}$ is of the form b/s with $b \in B$ integral over A and $s \in A \setminus \mathfrak{p}$, hence is integral

over $A_\mathfrak{p}$; in particular x satisfies an equation $x^n + c_1 x^{n-1} + \cdots + c_n = 0$ with $c_i \in A_\mathfrak{p}$ $(n \geq 1)$. Multiplying by a^{n-1} and using $ax = 1$:

$$x + c_1 + c_2 a + \cdots + c_n a^{n-1} = 0,$$

so $x \in A_\mathfrak{p}$. Then x and $a = x^{-1}$ are both in $A_\mathfrak{p}$, so x is a unit of $A_\mathfrak{p}$ and $B_\mathfrak{p} = xA_\mathfrak{p} = A_\mathfrak{p}$. Thus $(B/A)_\mathfrak{p} = B_\mathfrak{p}/A_\mathfrak{p} = 0$ for every prime $\mathfrak{p}$ (localization is exact, Cor. 2.11), and in particular for every maximal ideal. By Lemma 2.12, $B/A = 0$, i.e. $B = A$. □

Exercise 2.11

Problem. Let M be a flat A-module, and let $\rho : A \to B$ be a ring homomorphism. Show that the following properties are true.

(a) If A is an integral domain, then M is torsion-free.

(b) The canonical homomorphism $M \to M \otimes_A B$ is injective if and only if $\mathrm{Ker}(\rho)$ is contained in $\mathrm{Ann}(M)$.

Solution. (a) Let $a \in A$, $a \neq 0$. Since A is a domain, multiplication $t_a : A \to A$ is injective. As M is flat, $t_a \otimes \mathrm{id}_M : A \otimes_A M \to A \otimes_A M$ is injective; under $A \otimes_A M = M$ this is multiplication by a on M. So $ax = 0$ implies $x = 0$, i.e. M is torsion-free.

(b) Put $I = \mathrm{Ker}(\rho)$. The sequence $0 \to I \to A \to B$ is exact. Write it as $0 \to I \to A \to \rho(A) \to 0$ followed by the injection $\rho(A) \to B$. Tensoring with the flat module M gives an exact sequence $0 \to I \otimes M \to M \to \rho(A) \otimes M$ and an injection $\rho(A) \otimes M \to B \otimes M$. Hence

$$0 \to I \otimes_A M \to M \to M \otimes_A B$$

is exact, where the middle map is $x \mapsto x \otimes 1$. Therefore the kernel of $M \to M \otimes_A B$ is the image of $I \otimes_A M \to M$, which is IM. So the map is injective iff $IM = 0$, iff $ax = 0$ for all $a \in I$, $x \in M$, iff $I \subseteq \mathrm{Ann}(M)$. □

Exercise 2.12

Problem. Let B be a flat algebra over a Dedekind domain A. Let $f \in B$ be such that for every maximal ideal $\mathfrak{m}$ of A, the image of f in $B/\mathfrak{m}B$ is not a zero divisor. Show that B/fB is still flat over A.

Solution. By Cor. 2.14, a module over the Dedekind domain A is flat iff it is torsion-free. In particular B is torsion-free over A. We must show that B/fB is torsion-free.

Step 1: f is a non-zero-divisor on B/aB for every $a \in A$, $a \neq 0$. Let K be the kernel of multiplication by f on B/aB. Since localization is exact, $K_\mathfrak{m}$ is the kernel of f on $B_\mathfrak{m}/aB_\mathfrak{m}$, and $K = 0$ iff $K_\mathfrak{m} = 0$ for all maximal $\mathfrak{m}$ (Lemma 2.12). Fix $\mathfrak{m}$ and put $A' = A_\mathfrak{m}$, $B' = B_\mathfrak{m} = B \otimes_A A'$. If a is a unit of A' (in particular if A is a field), then $B'/aB' = 0$ and there is nothing to prove. Otherwise A' is a local principal ideal domain which is not a field, so its maximal ideal is $\pi A'$ with $\pi \neq 0$ and every nonzero element is a unit times a power of π; thus $aB' = \pi^e B'$ for some $e \geq 1$. Since B' is torsion-free over A' (flat), π is a non-zero-divisor on B'. Moreover

$B'/\pi B' = B \otimes_A A'/\mathfrak{m}A' = B/\mathfrak{m}B$ (elements of $A \setminus \mathfrak{m}$ already act invertibly on the $A/\mathfrak{m}$-module $B/\mathfrak{m}B$), and f is a non-zero-divisor there by hypothesis.

We show by induction on $e \geq 1$ that f is a non-zero-divisor on $B'/\pi^e B'$; the case $e = 1$ was just seen. Let $e \geq 2$ and let $x \in B'$ with $fx \in \pi^e B'$. Then $fx \in \pi B'$, so $x \in \pi B'$ by the case $e = 1$; write $x = \pi y$. Then $\pi f y = \pi^e z$ for some z, hence $fy = \pi^{e-1} z$ because π is a non-zero-divisor. By induction $y \in \pi^{e-1} B'$, so $x \in \pi^e B'$. This proves Step 1.

Step 2. Let $a \in A \setminus \{0\}$ and $b \in B$ with $ab \in fB$, say $ab = fc$. Then $fc \in aB$, so $c \in aB$ by Step 1, say $c = ac'$. Thus $a(b - fc') = 0$, and since B is torsion-free, $b = fc' \in fB$. So the class of b in B/fB is zero: B/fB is torsion-free, hence flat over A by Cor. 2.14. □

Exercise 2.13

> **Problem.** Let M be a faithfully flat A-module. Let $N' \to N \to N''$ be a sequence of A-modules. Show that it is exact if and only if the sequence $N' \otimes M \to N \otimes M \to N'' \otimes M$ is exact.

Solution. Write the maps as $u : N' \to N$, $v : N \to N''$.

($\Rightarrow$) This is flatness: if $\operatorname{Im} u = \operatorname{Ker} v$, then tensoring $0 \to \operatorname{Ker} v \to N \to \operatorname{Im} v \to 0$ and $N' \to \operatorname{Im} u \to 0$ with the flat module M shows that $\operatorname{Ker}(v \otimes 1) = \operatorname{Ker} v \otimes M = \operatorname{Im} u \otimes M = \operatorname{Im}(u \otimes 1)$ (all as submodules of $N \otimes M$, using that $\operatorname{Ker} v \otimes M \to N \otimes M$ is injective by flatness).

($\Leftarrow$) Assume the tensored sequence is exact. First, $vu = 0$. Indeed let $h = vu : N' \to N''$, $h = \iota \circ p$ with $p : N' \to \operatorname{Im} h$ onto and $\iota : \operatorname{Im} h \to N''$ the inclusion. Then $h \otimes 1 = (\iota \otimes 1)(p \otimes 1)$ and $(v \otimes 1)(u \otimes 1) = 0$ by assumption. Now $p \otimes 1$ is surjective, and $\iota \otimes 1$ is injective since M is flat. Hence $\operatorname{Im} h \otimes M = 0$, and so $\operatorname{Im} h = 0$ by Prop. 2.17(ii).

So $\operatorname{Im} u \subseteq \operatorname{Ker} v$; let $H = \operatorname{Ker} v / \operatorname{Im} u$. By flatness, $\operatorname{Ker} v \otimes M$ and $\operatorname{Im} u \otimes M$ are the submodules $\operatorname{Ker}(v \otimes 1)$ and $\operatorname{Im}(u \otimes 1)$ of $N \otimes M$ (as in the first paragraph), and by right exactness $H \otimes M = (\operatorname{Ker} v \otimes M)/(\operatorname{Im} u \otimes M) = \operatorname{Ker}(v \otimes 1)/\operatorname{Im}(u \otimes 1) = 0$. Hence $H = 0$ by Prop. 2.17(ii), i.e. the original sequence is exact. □

Exercise 2.14

> **Problem.** Let $A \to B$ be a ring homomorphism, and let J be an ideal of B such that B/J is flat over A. Show that for any ideal I of A, we have $(IB) \cap J = IJ$ (tensor the injection $I \to A$ by B/J).

Solution. The inclusion $IJ \subseteq IB \cap J$ is clear. Conversely let $x \in IB \cap J$ and write $x = \sum_k i_k b_k$ with $i_k \in I$, $b_k \in B$. Put $\xi = \sum_k i_k \otimes b_k \in I \otimes_A B$ and let $\xi' = \sum_k i_k \otimes \bar{b}_k \in I \otimes_A (B/J)$ be its image. Since B/J is flat, $I \otimes_A (B/J) \to A \otimes_A (B/J) = B/J$ is injective; it sends ξ' to $\sum i_k \bar{b}_k = \bar{x} = 0$ (as $x \in J$). Hence $\xi' = 0$.

By right exactness of $I \otimes_A -$ applied to $J \to B \to B/J \to 0$, the sequence $I \otimes_A J \to I \otimes_A B \to I \otimes_A (B/J) \to 0$ is exact, so ξ is in the image of $I \otimes_A J$: $\xi = \sum_l y_l \otimes c_l$ with $y_l \in I$, $c_l \in J$. Apply the multiplication map $I \otimes_A B \to B$, $i \otimes b \mapsto ib$: the image of ξ is $\sum_k i_k b_k = x$ and also $\sum_l y_l c_l \in IJ$. Thus $x \in IJ$. □

Exercise 2.15

Problem. Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of modules over a ring A. Let us suppose that M'' is flat. Show that M' is flat if and only if M is flat.

Solution. Let $N_0 \to N$ be an injective homomorphism of A-modules. Since M'' is flat, Prop. 2.6 gives commutative diagram with exact rows

$$0 \to M' \otimes N_0 \to M \otimes N_0 \to M'' \otimes N_0 \to 0$$
$$0 \to M' \otimes N \to M \otimes N \to M'' \otimes N \to 0$$

and vertical maps $\alpha', \alpha, \alpha''$ induced by $N_0 \to N$ (so each arrow of the first row is followed down by the corresponding vertical map). The map α'' is injective because M'' is flat.

(M flat $\Rightarrow M'$ flat.) Let $x \in M' \otimes N_0$ with $\alpha'(x) = 0$. Let y be the image of x in $M \otimes N_0$. Then $\alpha(y)$ is the image of $\alpha'(x) = 0$, so $\alpha(y) = 0$, and α is injective, so $y = 0$. Since $M' \otimes N_0 \to M \otimes N_0$ is injective (first row), $x = 0$.

(M' flat $\Rightarrow M$ flat.) Now α' and α'' are injective. Let $y \in M \otimes N_0$ with $\alpha(y) = 0$. The image of y in $M'' \otimes N_0$ is mapped by α'' to the image of $\alpha(y) = 0$, hence is 0; so y is the image of some $x \in M' \otimes N_0$. The image of $\alpha'(x)$ in $M \otimes N$ equals $\alpha(y) = 0$, and $M' \otimes N \to M \otimes N$ is injective (second row), so $\alpha'(x) = 0$, hence $x = 0$ and $y = 0$. So α is injective for every injection $N_0 \to N$: M is flat. □

Exercise 2.16

Problem. Let $(A, \mathfrak{m})$ be a Noetherian local ring, and $C^\bullet : 0 \to M' \to M \to M'' \to 0$ a complex of finitely generated flat A-modules. Show that if there exists an ideal $I \subseteq \mathfrak{m}$ such that $C^\bullet \otimes_A A/I$ is exact, then $C^\bullet$ is exact.

Solution. By Thm. 2.16, the finitely generated flat modules over the local ring A are free: $C^\bullet$ is $0 \to A^a \xrightarrow{\varphi} A^b \xrightarrow{\psi} A^c \to 0$ with $\psi\varphi = 0$. We use repeatedly: if N is a finitely generated A-module with $N = IN$, then $N = \mathfrak{m}N$ (as $I \subseteq \mathfrak{m}$), so $N = 0$ by Nakayama (Lemma 2.7). Also $N \otimes_A A/I = N/IN$.

ψ is surjective. $\operatorname{Coker}(\psi) \otimes A/I = \operatorname{Coker}(\psi \otimes A/I) = 0$ since $C^\bullet \otimes A/I$ is exact. So $\operatorname{Coker}\psi = I \operatorname{Coker}\psi$, and $\operatorname{Coker}\psi = 0$ by the remark above ($\operatorname{Coker}\psi$ is finitely generated). As A^c is free, ψ splits: $A^b \cong K \oplus A^c$ with $K = \operatorname{Ker}\psi$. In particular K is a direct summand of A^b, hence finitely generated and projective, and the sequence $0 \to K \to A^b \to A^c \to 0$ is split exact, so it stays exact after tensoring with A/I: $\operatorname{Ker}(\psi \otimes A/I) = K/IK$.

φ maps onto K. Since $\psi\varphi = 0$, φ factors as $\varphi : A^a \to K$. By exactness of $C^\bullet \otimes A/I$, the induced map $\bar{\varphi} : (A/I)^a \to \operatorname{Ker}(\psi \otimes A/I) = K/IK$ is bijective. Then $\operatorname{Coker}(\varphi) \otimes A/I = \operatorname{Coker}\bar{\varphi} = 0$, and $\operatorname{Coker}\varphi$ is finitely generated, so $\operatorname{Coker}\varphi = 0$.

φ is injective. As K is projective and $\varphi : A^a \to K$ is onto, it splits: $A^a \cong \operatorname{Ker}\varphi \oplus K$. Hence $\operatorname{Ker}\varphi$ is finitely generated and $0 \to \operatorname{Ker}\varphi \to A^a \to K \to 0$ stays exact after $\otimes A/I$, so $\operatorname{Ker}\varphi \otimes A/I = \operatorname{Ker}\bar{\varphi} = 0$. By the remark, $\operatorname{Ker}\varphi = 0$.

Thus φ is injective, $\operatorname{Im}\varphi = K = \operatorname{Ker}\psi$, and ψ is surjective: $C^\bullet$ is exact. $\square$

Exercise 2.17

Problem. Let M be a finitely generated flat A-module. Show that it is faithfully flat if and only if $\operatorname{Ann}(M) = 0$.

Solution. ($\Rightarrow$) Let $J = \operatorname{Ann}(M)$ and $\iota : J \to A$ the inclusion. Since M is flat, $\iota \otimes 1 : J \otimes_A M \to A \otimes_A M = M$ is injective; its image is $JM = 0$. So $J \otimes_A M = 0$, and then $J = 0$ by Prop. 2.17(ii).

($\Leftarrow$) Suppose $\operatorname{Ann}(M) = 0$; we verify Prop. 2.17(i). Let $\mathfrak{m}$ be a maximal ideal of A and suppose $M = \mathfrak{m}M$. Localizing, $M_\mathfrak{m} = \mathfrak{m}M_\mathfrak{m}$ with $M_\mathfrak{m}$ finitely generated over the local ring $A_\mathfrak{m}$, so $M_\mathfrak{m} = 0$ by Nakayama (Lemma 2.7). If $x_1, \ldots, x_n$ generate M, each $x_i/1 = 0$ in $M_\mathfrak{m}$, so there are $s_i \in A \setminus \mathfrak{m}$ with $s_i x_i = 0$. Then $s = s_1 \cdots s_n \in A \setminus \mathfrak{m}$ kills M, i.e. $s \in \operatorname{Ann}(M) = 0$. But $0 \in \mathfrak{m}$, a contradiction. Hence $M \neq \mathfrak{m}M$ for all maximal $\mathfrak{m}$, and M is faithfully flat. $\square$

Exercise 2.18

Problem. Let B be an A-algebra, and let E be a faithfully flat B-module. Show that E is flat over A if and only if B is flat over A.

Solution. ($\Leftarrow$) E is flat over B and B is flat over A, so E is flat over A by Prop. 2.2(d).

($\Rightarrow$) Assume E is flat over A. Let $N' \to N$ be an injective homomorphism of A-modules and let $g : N' \otimes_A B \to N \otimes_A B$ be the induced B-linear map; we must show g injective. Using the natural isomorphisms $(N' \otimes_A B) \otimes_B E \cong N' \otimes_A E$ (and the same for N), the map $g \otimes_B \mathrm{id}_E$ becomes $N' \otimes_A E \to N \otimes_A E$, which is injective since E is flat over A. Let $L = \operatorname{Ker} g$. Since E is flat over B, $L \otimes_B E = \operatorname{Ker}(g \otimes_B \mathrm{id}_E) = 0$. As E is faithfully flat over B, $L = 0$ by Prop. 2.17(ii). So g is injective and B is flat over A. $\square$

Exercise 2.19

Problem. Let $f : A \to B$ be a faithfully flat ring homomorphism.

(a) Show that f is injective and that $I \to I \otimes_A B$ is injective for every ideal I of A.

(b) Let $N = \operatorname{Coker}(f)$ be the cokernel of f. Let I be an ideal of A. Using the commutative diagram

$$\begin{array}{l} I \to I \otimes_A B \to I \otimes_A N \to 0 \\ 0 \to I \to IB \to IN \to 0 \end{array}$$

and Exercise 2.6(b), show that $I \otimes_A N \to IN$ is injective, and hence N is a flat A-module.

(c) Show that for any A-module M, the canonical map $M \to M \otimes_A B$ is injective.

Solution. (a) Let $K = \operatorname{Ker} f$. Since B is flat, $K \otimes_A B \to A \otimes_A B = B$ is injective, and it sends $k \otimes b$ to $f(k)b = 0$. Hence $K \otimes_A B = 0$ and $K = 0$ by Prop. 2.17(ii): f is injective. Now let I be an ideal of A. The composite $I \to I \otimes_A B \to B$, $x \mapsto x \otimes 1 \mapsto f(x)$, is injective since f is; hence $I \to I \otimes_A B$ is injective.

(b) We first record Ex. 2.6(b): $IB \cap A = I$ (identifying A with $f(A) \subseteq B$, allowed by (a)). Indeed let $P = (IB \cap A)/I$, the kernel of $A/I \to B/IB$. By flatness of B, $0 \to P \otimes_A B \to (A/I) \otimes_A B \to (B/IB) \otimes_A B$ is exact. The composite $B/IB \cong (A/I) \otimes_A B \to (B/IB) \otimes_A B$ sends $\bar{b}$ to $\bar{1} \otimes b$; it has a left inverse $\bar{c} \otimes b \mapsto \overline{cb}$, so it is injective. Hence $P \otimes_A B = 0$ and $P = 0$, i.e. $IB \cap A = I$.

Identify A with a subring of B, so $0 \to A \to B \to N \to 0$ is exact. Then $IN = (IB + A)/A \cong IB/(IB \cap A) = IB/I$, so the bottom row $0 \to I \to IB \to IN \to 0$ is exact. The top row $I \to I \otimes_A B \to I \otimes_A N \to 0$ is exact by right exactness of $I \otimes_A -$. The vertical maps are the identity of I, the map $I \otimes_A B \to IB$, $x \otimes b \mapsto xb$, and the map $I \otimes_A N \to IN$, $x \otimes \bar{b} \mapsto x\bar{b}$; the diagram commutes, and the middle vertical map $I \otimes_A B \to IB$ is an isomorphism because B is flat (it is the injection $I \otimes B \to B$ with image IB).

Let $z \in I \otimes_A N$ map to 0 in IN. Lift z to $w \in I \otimes_A B$. The image of w in IB maps to 0 in IN, hence lies in (the image of) I: it is the image of some $x \in I$. Then w and the image w_x of x in $I \otimes_A B$ have the same image in IB, so $w = w_x$ by injectivity of the middle map. Therefore z is the image of w_x in $I \otimes_A N$, which is 0 since two consecutive maps of the top row compose to zero. Thus $I \otimes_A N \to IN$ is injective. Composed with $IN \subseteq N$ this says that $I \otimes_A N \to A \otimes_A N = N$ is injective for every ideal I, so N is flat by Thm. 2.4.

(c) Let M be an A-module. The sequence $0 \to A \xrightarrow{f} B \to N \to 0$ is exact and N is flat by (b), so by Prop. 2.6 the sequence $0 \to M \otimes_A A \to M \otimes_A B \to M \otimes_A N \to 0$ is exact. In particular $M = M \otimes_A A \to M \otimes_A B$, $x \mapsto x \otimes 1$, is injective. □

1.3 Formal completion

Conventions: rings are commutative with 1; "complete" means separated and complete, as in the text (so A is complete for the I-adic topology iff the canonical map $A \to \widehat{A}$ is bijective). For a topological group G given by a filtration (G_n), the closure of a subgroup H is $\bigcap_n (H + G_n)$.

Exercise 3.1

Problem. Is the usual topology on $\mathbf{R}$ defined by a subgroup filtration?

Solution. No. Suppose $(G_n)_n$ is a descending chain of subgroups of $\mathbf{R}$ defining the usual topology, so the G_n are neighborhoods of 0 for the usual topology. Then $G_n \supseteq (-\varepsilon, \varepsilon)$ for some $\varepsilon > 0$. For $x \in \mathbf{R}$ choose an integer $k \geq 1$ with $|x/k| < \varepsilon$; then $x = k \cdot (x/k) \in G_n$ because G_n is a group. Hence $G_n = \mathbf{R}$ for all n. But then the only neighborhood of 0 for the topology defined by the filtration is $\mathbf{R}$ itself (the topology is indiscrete), whereas $(-1, 1)$ is a neighborhood of 0 for the usual topology. Contradiction. (The argument shows that no family of subgroups at all can give a fundamental system of neighborhoods of 0 for the usual topology.) □

Exercise 3.2

Problem. Show that $\mathbf{Z}_p$ is a principal local ring, with maximal ideal $p\mathbf{Z}_p$.

Solution. Recall $\mathbf{Z}_p = \varprojlim \mathbf{Z}/p^n\mathbf{Z}$ with $\widehat{G}_n = \{a = (a_m) \in \mathbf{Z}_p : a_m = 0 \text{ for } m \leq n\} = \mathrm{Ker}(\mathbf{Z}_p \to \mathbf{Z}/p^n\mathbf{Z})$.

Step 1: $\widehat{G}_n = p^n\mathbf{Z}_p$. Clearly $p^n\mathbf{Z}_p \subseteq \widehat{G}_n$. Conversely let $a \in \widehat{G}_n$. For $m \geq n$, $a_m \in \mathbf{Z}/p^m\mathbf{Z}$ reduces to 0 modulo p^n, so $a_m = p^n b_m$ for a unique $b_m \in \mathbf{Z}/p^{m-n}\mathbf{Z}$ (multiplication by p^n is an injection $\mathbf{Z}/p^{m-n}\mathbf{Z} \to \mathbf{Z}/p^m\mathbf{Z}$ with image $p^n\mathbf{Z}/p^m\mathbf{Z}$). Uniqueness shows that $b_{m+1} \mapsto b_m$, so the family $b'_k = b_{k+n} \in \mathbf{Z}/p^k\mathbf{Z}$ $(k \geq 1)$ is an element $b' \in \mathbf{Z}_p$, and $p^n b' = a$ componentwise: the k-th component of $p^n b'$ is $p^n b_{k+n}$ reduced modulo p^k, which is the reduction of a_{k+n} modulo p^k, namely a_k. Thus $\mathbf{Z}_p/p^n\mathbf{Z}_p \simeq \mathbf{Z}/p^n\mathbf{Z}$ and the topology of $\mathbf{Z}_p$ is the p-adic one; in particular $\bigcap_n p^n\mathbf{Z}_p = 0$ since $\bigcap_n \widehat{G}_n = 0$.

Step 2: units. If $u \in \mathbf{Z}_p \setminus p\mathbf{Z}_p$, then $u_1 \neq 0$ in $\mathbf{Z}/p\mathbf{Z}$, so each u_m is a unit of $\mathbf{Z}/p^m\mathbf{Z}$ (it is nonzero modulo p). The inverses u_m^{-1} are unique, hence compatible, and define an inverse of u in $\mathbf{Z}_p$. Conversely elements of $p\mathbf{Z}_p$ are not units (their image in $\mathbf{Z}/p\mathbf{Z}$ is 0 and $1 \neq 0$ there). So $\mathbf{Z}_p^\times = \mathbf{Z}_p \setminus p\mathbf{Z}_p$.

Step 3. Let $a \neq 0$. Since $a \in p^0\mathbf{Z}_p$ and $\bigcap_n p^n\mathbf{Z}_p = 0$, there is a largest n with $a \in p^n\mathbf{Z}_p$; write $a = p^n u$, and $u \notin p\mathbf{Z}_p$ by maximality, so u is a unit. Also $p^k \neq 0$ in $\mathbf{Z}_p$ for every k (its image in $\mathbf{Z}/p^{k+1}\mathbf{Z}$ is nonzero), hence $p^k w \neq 0$ for each unit w. Therefore $\mathbf{Z}_p$ is an integral domain: $p^n u \cdot p^m v = p^{n+m} uv \neq 0$.

Let $J \neq 0$ be an ideal and let n be the least integer such that J contains an element $p^n u$ with u a unit (every nonzero element has this form). Then $p^n \in J$, and every element of J is $p^k w$ with $k \geq n$, so $J = p^n\mathbf{Z}_p$. Hence $\mathbf{Z}_p$ is a principal ideal domain. The non-units are exactly $p\mathbf{Z}_p$ (Step 2), an ideal, so $\mathbf{Z}_p$ is local with maximal ideal $p\mathbf{Z}_p$. □

Exercise 3.3

Problem. Let A be the ring of germs of real analytic functions in $0 \in \mathbf{R}$. Let $x \in A$ be the identity map on $\mathbf{R}$. Show that the map $f \mapsto \sum_n (f^{(n)}(0)/n!)T^n$ from A to $\mathbf{R}[[T]]$ is injective, and induces an isomorphism from the xA-adic completion of A onto $\mathbf{R}[[T]]$. Show that this is false if we replace A by the C^∞-functions in 0.

Solution. Write $\tau : A \to \mathbf{R}[[T]]$ for the Taylor map. It is a ring homomorphism (a germ of analytic function is, near 0, the sum of its Taylor series, and the product of convergent series is the Cauchy product).

Injective. If $\tau(f) = 0$ then f equals its Taylor series 0 on a neighborhood of 0, so the germ is 0.

Powers of xA. We claim $x^n A = \{f : \tau(f) \in T^n\mathbf{R}[[T]]\}$. The inclusion $\subseteq$ is clear. If $f = \sum_{k \geq n} a_k x^k$ converges on $(-r, r)$, then $g = \sum_{k \geq 0} a_{k+n} x^k$ has the same radius of convergence, so it is a germ in A, and $f = x^n g$. Consequently τ induces an isomorphism $A/x^n A \to \mathbf{R}[[T]]/T^n\mathbf{R}[[T]] = \mathbf{R}[T]/(T^n)$: it is injective by the claim, and surjective since polynomials lie in A. These isomorphisms are compatible with the transition maps, so they induce an isomorphism

$$\widehat{A} = \varprojlim A/x^n A \xrightarrow{\sim} \varprojlim \mathbf{R}[[T]]/T^n\mathbf{R}[[T]] = \mathbf{R}[[T]],$$

the last equality because a compatible family of classes mod T^n is exactly a power series (Example 3.6 with $A = \mathbf{R}$, $r = 1$). The composite $A \to \widehat{A} \to \mathbf{R}[[T]]$ is τ, so the isomorphism is the one induced by τ.

The C^∞ case. Let E be the ring of germs at 0 of C^∞ functions. The Taylor map $E \to \mathbf{R}[[T]]$ is *not* injective: $f_0(x) = e^{-1/x^2}$ ($f_0(0) = 0$) is C^∞, all its derivatives vanish at 0, but it is not the zero germ. So the statement "$f \mapsto \tau(f)$ is injective" is false. Moreover, for $f \in E$ with vanishing derivatives of order $< n$, Taylor's formula with integral remainder $f(x) = x^n \int_0^1 \frac{(1-t)^{n-1}}{(n-1)!} f^{(n)}(tx)\, dt$ shows that $f = x^n g$ with $g \in E$; hence again $E/x^n E \simeq \mathbf{R}[T]/(T^n)$ and the xE-adic completion is still $\mathbf{R}[[T]]$, but the canonical map $E \to \widehat{E} = \mathbf{R}[[T]]$ has kernel $\bigcap_n x^n E \ni f_0$ (it is surjective by Borel's lemma). So E is not separated for the xE-adic topology, and $E \to \widehat{E}$ is not an isomorphism onto its image, in contrast with the analytic case. □

Exercise 3.4

Problem. Find examples where $M \otimes_A \widehat{A} \to \widehat{M}$ is not surjective, or is not injective.

Solution. Take $A = \mathbf{Z}$, $I = p\mathbf{Z}$, so $\widehat{A} = \mathbf{Z}_p$ (Ex. 3.2). By Lemma 3.9 and Cor. 3.14 the map is surjective, resp. bijective, for finitely generated M, so M must be infinitely generated.

Not surjective. Let $M = \bigoplus_{k\geq 1} \mathbf{Z}e_k$. Then $\widehat{M} = \varprojlim M/p^n M$ with $M/p^n M = \bigoplus_k (\mathbf{Z}/p^n)e_k$. Put $x_n = \sum_{k=1}^n p^{k-1} e_k \in M/p^n M$. Since $x_{n+1} - x_n \equiv p^n e_{n+1} \equiv 0 \pmod{p^n}$, we get $x = (x_n) \in \widehat{M}$. The image of $M \otimes \mathbf{Z}_p = \bigoplus_k \mathbf{Z}_p e_k$ consists of elements $\sum_{k\leq N} a_k e_k$ (finite sums, $a_k \in \mathbf{Z}_p$) whose n-th component has e_{N+1}-coordinate 0 for all n. But the e_{N+1}-coordinate of x_n is $p^N \neq 0$ in $\mathbf{Z}/p^n$ for $n > N$. So x is not in the image.

Not injective. Let $M = \mathbf{Q}$. Then $p^n \mathbf{Q} = \mathbf{Q}$ for all n, so $\widehat{M} = \varprojlim \mathbf{Q}/\mathbf{Q} = 0$. But $\mathbf{Q} \otimes_{\mathbf{Z}} \mathbf{Z}_p = S^{-1}\mathbf{Z}_p$ with $S = \mathbf{Z} \setminus \{0\}$; the map $\mathbf{Z} \to \mathbf{Z}_p$ is injective (a nonzero integer m has nonzero image in $\mathbf{Z}/p^k\mathbf{Z}$ for k large), so the image of S in the domain $\mathbf{Z}_p$ (Ex. 3.2) does not contain 0, and hence $\mathbf{Z}_p$ embeds in $S^{-1}\mathbf{Z}_p$; in particular $S^{-1}\mathbf{Z}_p \neq 0$. So the map $\mathbf{Q} \otimes \mathbf{Z}_p \to \widehat{\mathbf{Q}} = 0$ is not injective. □

Exercise 3.5

Problem. Let $B = \bigoplus_{n\geq 0} B_n$ be a graded ring. (a) Show that B_0 is a subring of B, with unit. (b) Show that if B is Noetherian, then so is B_0, and B is a finitely generated algebra over B_0.

Solution. (a) B_0 is an additive subgroup and $B_0 B_0 \subseteq B_0$. Write $1 = \sum_d e_d$ with $e_d \in B_d$. For $b \in B_n$, the equation $b = b \cdot 1 = \sum_d b e_d$ has $be_d \in B_{n+d}$; comparing components of degree n in the direct sum gives $b = be_0$. Since B is generated additively by homogeneous elements, e_0 acts as the identity on B, so $e_0 = e_0 \cdot 1 = 1$. Thus $1 \in B_0$ and B_0 is a subring with the same unit.

(b) Let $B_+ = \bigoplus_{n\geq 1} B_n$. It is an ideal (as $B_d B_e \subseteq B_{d+e}$) and $B = B_0 \oplus B_+$, so the projection $B \to B_0$ is a surjective ring homomorphism with kernel B_+, i.e. $B_0 \simeq B/B_+$. Hence B_0 is Noetherian when B is.

The ideal B_+ is generated by the set of homogeneous elements of degree ≥ 1 (these span B_+ additively). Since B is Noetherian, finitely many of them, say $x_1, \ldots, x_r$ of degrees $d_i \geq 1$,

already generate B_+. Let $C = B_0[x_1, \ldots, x_r] \subseteq B$. We show $B_n \subseteq C$ by induction on n. For $n = 0$ it is clear. Let $n \geq 1$ and $b \in B_n \subseteq B_+$; write $b = \sum_i c_i x_i$ with $c_i \in B$. Taking the degree-n component, $b = \sum_i (c_i)_{n-d_i} x_i$, where $(c_i)_{n-d_i} \in B_{n-d_i}$ (interpreted as 0 if $n < d_i$). Since $n - d_i < n$, these lie in C by induction, so $b \in C$. Hence $B = C$ is a finitely generated B_0-algebra. □

Exercise 3.6

> **Problem.** Let A be an integral domain that is not a field, and let K be its field of fractions. Show that the conclusion of Prop. 3.11 does not hold if we take $M = K$, $N = A$, and $I \neq 0$.

Solution. (The printed hypothesis "$I \neq 0$" must be read as: I is a nonzero *proper* ideal; such ideals exist since A is not a field. For $I = A$ the conclusion holds trivially, and for $I = 0$ as well, so the restriction is necessary.)

Put $M_n = I^n K$. Choose $0 \neq a \in I$. For any $y \in K$ we have $y = a^n \cdot (y/a^n) \in I^n K$, so $M_n = K$ for all n. The filtration (M_n) is an I-filtration ($IM_n \subseteq M_{n+1}$) and it is stable: $M_{n+1} = K = IM_n$. Now $N = A$ and $M_n \cap N = A$ for all n. The filtration $(M_n \cap N)_n$ is stable iff $A = I \cdot A = I$ for $n \gg 0$, which fails because $I \neq A$. So the conclusion of Prop. 3.11 fails. (The hypotheses "A Noetherian, M finitely generated" of Prop. 3.11 are both violated: K is not finitely generated over A when A is Noetherian and not a field, and A need not be Noetherian.) □

Exercise 3.7

> **Problem.** Let $(A, \mathfrak{m})$ be a Noetherian local ring. Show that $\bigcap_{n \geq 0} \mathfrak{m}^n = 0$. Give a counter-example with A not Noetherian.

Solution. By Cor. 3.13 (with $M = A$, $I = \mathfrak{m}$), $\bigcap_n \mathfrak{m}^n$ is the set of x for which there is $\alpha \in \mathfrak{m}$ with $(1 + \alpha)x = 0$. In a local ring $1 + \alpha$ is a unit for $\alpha \in \mathfrak{m}$ (otherwise $1 + \alpha \in \mathfrak{m}$ and so $1 \in \mathfrak{m}$). Hence $x = 0$.

Counter-example. Let E be the ring of germs at 0 of C^∞ functions on $\mathbf{R}$ (Ex. 3.3). If $f(0) \neq 0$ then $1/f$ is C^∞ near 0, so f is a unit; hence E is local with maximal ideal $\mathfrak{m} = \{f : f(0) = 0\}$. By Taylor's formula in Ex. 3.3, $\mathfrak{m} = xE$ and $\mathfrak{m}^n = x^n E$. The germ $f_0 = e^{-1/x^2}$ lies in every $\mathfrak{m}^n$ (indeed f_0/x^n, extended by 0, is C^∞) and is nonzero. So $\bigcap_n \mathfrak{m}^n \neq 0$; E is not Noetherian by the first part. □

Exercise 3.8

> **Problem.** Let A be a Noetherian ring, and I, J ideals of A. Let $\widehat{A}$ be the I-adic completion of A and $(A/J)^\wedge$ the completion of A/J for the $(I + J)/J$-adic topology. Show that there is a canonical isomorphism $\widehat{A}/J\widehat{A} \simeq (A/J)^\wedge$.

Solution. Let $M = A/J$, a finitely generated A-module. For $n \geq 0$ we have $I^n M = (I^n + J)/J = ((I + J)/J)^n$ (the n-th power of the image of I in A/J is the image of I^n, and $(I + J)/J$ is the

image of I). So the I-adic topology of the A-module M is the $(I+J)/J$-adic topology of the ring A/J, and the I-adic completion $\widehat{M}$ of the module M is $(A/J)^\wedge$. By Cor. 3.14, the canonical map $M \otimes_A \widehat{A} \to \widehat{M}$ is an isomorphism. Finally

$$M \otimes_A \widehat{A} = (A/J) \otimes_A \widehat{A} \simeq \widehat{A}/J\widehat{A}.$$

Composing gives the canonical isomorphism $\widehat{A}/J\widehat{A} \simeq (A/J)^\wedge$ (it sends the class of $\alpha = (\alpha_n)_n$ to the family of the images of α_n in $A/(I^n+J)$). □

Exercise 3.9

> **Problem.** (*nth root*) Let $n \geq 2$ be an integer. Let $D = \mathbf{Z}[1/n]$. (a) Consider the polynomial $S = (1+T)^n - 1 \in D[T]$. Show that $D[[S]] = D[[T]]$ and that there exists an $f(S) \in SD[[S]]$ such that $1 + S = (1 + f(S))^n$. (b) Let A be a complete ring for the I-adic topology, where I is an ideal of A. Suppose that n is invertible in A. Let $x \in I$. Show that there exists a unique continuous homomorphism $\varphi : D[[S]] \to A$ such that $\varphi(S) = x$. Conclude that there exists a $y \in I$ such that $1 + x = (1 + y)^n$. (c) Show, by giving an example, that statement (b) is false if n is not invertible in A.

Solution. Lemma L. Let R be a ring and $g = c_1T + c_2T^2 + \cdots \in R[[T]]$ with $c_1 \in R^\times$. Then the substitution $\sigma : R[[X]] \to R[[T]]$, $\sum b_k X^k \mapsto \sum b_k g^k$ (a T-adically convergent sum, since $g^k \in T^k R[[T]]$) is a ring isomorphism.

Proof. σ is a ring homomorphism (it is the substitution homomorphism). We have $g^k = c_1^k T^k +$ (higher terms). If $h = \sum b_k X^k \neq 0$ with lowest nonzero coefficient b_m, then $\sigma(h)$ has T^m-coefficient $b_m c_1^m \neq 0$ and no lower terms; so σ is injective. For surjectivity let $H \in R[[T]]$. Construct $b_0, b_1, \ldots$ inductively so that $H - \sum_{k<m} b_k g^k \in T^m R[[T]]$: this is true for $m = 0$; if it holds for m, let e be the T^m-coefficient of $H - \sum_{k<m} b_k g^k$ and put $b_m = e c_1^{-m}$; then $H - \sum_{k\leq m} b_k g^k \in T^{m+1} R[[T]]$. Then $\sigma(\sum b_k X^k) - H \in \bigcap_m T^m R[[T]] = 0$. (Lemma)

(a) Write $S = nT + \binom{n}{2} T^2 + \cdots + T^n$. Its coefficient of T is n, a unit of D. By Lemma L (with $R = D$, $g = S$), $\sigma : D[[X]] \to D[[T]]$, $X \mapsto S$, is an isomorphism, which is the meaning of $D[[S]] = D[[T]]$. Let $f(X) = \sigma^{-1}(T) \in D[[X]]$. Its constant term equals the constant term of $\sigma(f) = T$, namely 0, so $f \in XD[[X]]$, i.e. $f(S) \in SD[[S]]$. Moreover $\sigma((1+f)^n) = (1+T)^n = 1 + S = \sigma(1+X)$, and σ is injective, so $(1 + f(X))^n = 1 + X$ in $D[[X]]$.

(b) *Existence.* Since n is invertible in A, the ring map $\mathbf{Z} \to A$ extends uniquely to $D \to A$; we regard $D \to A$ as given. For $g = \sum b_k S^k \in D[[S]]$ put $\varphi_N(g) = \sum_{k<N} b_k x^k$. Since $x^k \in I^k$, $(\varphi_N(g))_N$ is a Cauchy sequence; as A is complete (so separated) it has a unique limit $\varphi(g)$. Clearly φ is additive and $\varphi(S) = x$. For $g = \sum b_k S^k$, $h = \sum c_k S^k$, $\varphi_N(gh) - \varphi_N(g)\varphi_N(h) = -\sum_{i,j<N,\ i+j\geq N} b_i c_j x^{i+j} \in I^N$; letting $N \to \infty$ gives $\varphi(gh) = \varphi(g)\varphi(h)$ (limits are unique and the ring operations are continuous). Also $\varphi(1) = 1$. Finally $\varphi(S^N g) = x^N \varphi(g) \in I^N$, so $\varphi(S^N D[[S]]) \subseteq I^N$ and φ is continuous.

Uniqueness. Let ψ be any ring homomorphism with $\psi(S) = x$ (continuity is not even needed). On D it is the canonical map, since $\psi(n) = n$ is invertible, so $\psi(1/n) = n^{-1}$. Write $g = \sum_{k<N} b_k S^k + S^N g_N$; then $\psi(g) - \varphi_N(g) = x^N \psi(g_N) \in I^N$. The same holds for φ, so $\psi(g) - \varphi(g) \in \bigcap_N I^N = 0$ because A is separated.

The root. By (a), $f(S) = Sf_1$ with $f_1 \in D[[S]]$. Put $y = \varphi(f(S)) = x\varphi(f_1) \in I$. Applying φ to $(1 + f(S))^n = 1 + S$ gives $(1 + y)^n = 1 + x$.

(c) Take $n = 2$, $A = \mathbf{Z}/8\mathbf{Z}$, $I = 2A$. Then $I^3 = 0$, so A is complete and separated for the I-adic topology ($\widehat{A} = \varprojlim A/I^k = A$), but 2 is not invertible in A. Take $x = 2 \in I$. If $3 = 1+x = (1+y)^2$ with $y \in I$, then $1 + y$ is odd and its square is $\equiv 1 \pmod 8$ (odd squares are 1 mod 8), whereas $3 \not\equiv 1 \pmod 8$. So no such y exists. □

Exercise 3.10

> **Problem.** Let k be a field of characteristic different from 2, and $A = k[x, y]/(y^2 - x^2(x+1))$. Let $\widehat{A}$ be the $\mathfrak{m}$-adic completion, where $\mathfrak{m} = (x, y)A$. Show that $\widehat{A} \simeq k[[u, v]]/(uv)$.

Solution. Let $B = k[x, y]$, $\mathfrak{n} = (x, y)$, $F = y^2 - x^2(x + 1)$. By Example 3.6, $\widehat{B} = k[[x, y]]$ ($\mathfrak{n}$-adic). Apply Ex. 3.8 with $I = \mathfrak{n}$, $J = (F)$ (B is Noetherian): the $(I + J)/J$-adic topology on $A = B/J$ is the $\mathfrak{m}$-adic one, so

$$\widehat{A} \simeq k[[x, y]]/F\,k[[x, y]].$$

Square root of $1 + x$. The ring $k[[x]] = \widehat{k[x]}$ is complete for the (x)-adic topology and 2 is invertible in it (char $k \neq 2$). By Ex. 3.9(b) (applied to $A = k[[x]]$, $I = (x)$, $n = 2$, and the element $x \in I$) there is $w = 1 + y_0 \in k[[x]]$, $y_0 \in (x)$, with $w^2 = 1 + x$. Then w is a unit, $w = 1 +$ (terms of degree ≥ 1) and

$$F = y^2 - x^2w^2 = (y - xw)(y + xw) \quad \text{in } k[[x, y]].$$

Change of variables. Put $u = y - xw$, $v = y + xw$, elements of $\mathfrak{n}k[[x, y]]$. We show that $k[[x, y]]$ is identified with $k[[u, v]]$ so that u, v become the variables. Use the standard universal property (same proof as in Ex. 3.9(b)): for a ring R complete for the $\mathfrak{J}$-adic topology and elements $r, s \in \mathfrak{J}$ there is a unique ring homomorphism $k[[U, V]] \to R$ with $U \mapsto r, V \mapsto s$ extending the given map from k.

Let $g = xw \in k[[x]]$; its coefficient of x is 1, so by Lemma L (Ex. 3.9), $\sigma_g : k[[X]] \to k[[x]]$, $X \mapsto g$, is an isomorphism; let $h = \sigma_g^{-1}(x) \in k[[X]]$, so $h(g(x)) = x$, and also $g(h(X)) = X$ (indeed $\sigma_g(g \circ h) = g \circ h \circ g = g \circ x = g = \sigma_g(X)$, and σ_g is injective). Define

$$\Phi : k[[U, V]] \to k[[x, y]],\ U \mapsto u,\ V \mapsto v; \qquad \Psi : k[[x, y]] \to k[[U, V]],\ x \mapsto h\big(\tfrac{V-U}{2}\big),\ y \mapsto \tfrac{U+V}{2},$$

legitimate since 2 is invertible and the images lie in the maximal ideal (U, V), resp. (x, y). Then, with $t = (V - U)/2$, $\Psi(u) = \frac{U+V}{2} - g(h(t)) = \frac{U+V}{2} - t = U$ and likewise $\Psi(v) = \frac{U+V}{2} + t = V$. So $\Psi\Phi$ and the identity are homomorphisms $k[[U, V]] \to k[[U, V]]$ agreeing on U, V, hence equal by uniqueness. Conversely $\Phi\Psi(y) = \frac{u+v}{2} = y$, and $\Phi\Psi(x) = h\big(\frac{v-u}{2}\big) = h(xw) = h(g(x)) = x$; so $\Phi\Psi = \mathrm{id}$ by uniqueness. Hence Φ is an isomorphism, with $\Phi(UV) = uv = F$. Therefore

$$\widehat{A} \simeq k[[x, y]]/(F) \simeq k[[U, V]]/(UV).$$

□

Exercise 3.11

Problem. Let A be a complete ring for the I-adic topology, where I is an ideal of A. Let $(M_n)_{n\geq 0}$ be A-modules such that $I^{n+1}M_n = 0$ and that there exist surjective homomorphisms $\pi_n : M_{n+1} \to M_n$ with $\mathrm{Ker}(\pi_n) = I^{n+1}M_{n+1}$. Let $M = \varprojlim_n M_n$ and denote the (surjective) canonical homomorphisms by $u_n : M \to M_n$.

(a) Fix $d \geq 0$. Show that for any $n \geq d$, there is an exact sequence $0 \to I^{d+1}M_n \to M_n \to M_d \to 0$.

(b) Suppose M_0 is generated over A by a finite number of elements $e_{0,1}, \ldots, e_{0,m}$. Let $e_1, \ldots, e_m \in M$ be such that $u_0(e_i) = e_{0,i}$, and define $\varphi_n : A^m \to M_n$ by $(a_1, \ldots, a_m) \mapsto \sum_i a_i u_n(e_i)$. Show that φ_n is surjective, and, using Lemma 3.1, that M is generated by the e_i.

(c) Suppose moreover that A is Noetherian and M_0 is finitely generated over A. Show that $\operatorname{Ker} u_n = I^{n+1}M$ (in other words, u_n induces an isomorphism $M/I^{n+1}M \simeq M_n$). In (c), the book proceeds in three steps: (1) Let $K_n = \operatorname{Ker} u_n$. Show that $K_n = I^{n+1}M + K_{n+1}$. (2) Show that the canonical homomorphism $\varprojlim_n M/I^{n+1}M \to \varprojlim_n M_n$ is an isomorphism (use Cor. 3.14). (3) Apply Lemma 3.1 to the exact sequence of inverse systems $0 \to (K_n/I^{n+1}M)_n \to (M/I^{n+1}M)_n \to (M_n)_n \to 0$ and show that $K_n = I^{n+1}M$.

Solution. (Parts (1)–(3) are written out as steps of (c).) Throughout, $\pi_{n,d} : M_n \to M_d$ $(n \geq d)$ denotes the composite $\pi_d \circ \cdots \circ \pi_{n-1}$, so $\pi_d \circ u_{d+1} = u_d$ etc. The maps u_n are surjective because the π_n are (lift an element of M_n successively to $M_{n+1}, M_{n+2}, \ldots$).

(a) $\pi_{n,d}$ is surjective as a composite of surjections. We show by induction on $n \geq d$ that $\operatorname{Ker} \pi_{n,d} = I^{d+1}M_n$. For $n = d$, $\pi_{d,d} = \mathrm{id}$ and $I^{d+1}M_d = 0$ by hypothesis. Suppose it holds for n. Then

$$\operatorname{Ker} \pi_{n+1,d} = \pi_n^{-1}(\operatorname{Ker} \pi_{n,d}) = \pi_n^{-1}(I^{d+1}M_n) = I^{d+1}M_{n+1} + \operatorname{Ker} \pi_n,$$

the last equality because π_n is surjective, so $\pi_n(I^{d+1}M_{n+1}) = I^{d+1}M_n$. Since $\operatorname{Ker} \pi_n = I^{n+1}M_{n+1} \subseteq I^{d+1}M_{n+1}$ for $n \geq d$, the right side is $I^{d+1}M_{n+1}$. This proves the exact sequence $0 \to I^{d+1}M_n \to M_n \xrightarrow{\pi_{n,d}} M_d \to 0$.

(b) *Surjectivity of φ_n,* by induction on n. For $n = 0$, φ_0 is onto since $u_0(e_i) = e_{0,i}$ generate M_0. Suppose φ_n is onto and let $N = \varphi_{n+1}(A^m) \subseteq M_{n+1}$. Since $\pi_n \circ u_{n+1} = u_n$, $\pi_n(N) = \varphi_n(A^m) = M_n$, so $M_{n+1} = N + \operatorname{Ker} \pi_n = N + I^{n+1}M_{n+1}$. Substituting this into itself, $I^{n+1}M_{n+1} = I^{n+1}N + I^{2n+2}M_{n+1}$, and $I^{2n+2}M_{n+1} \subseteq I^{n+2}M_{n+1} = 0$. Hence $M_{n+1} = N + I^{n+1}N = N$.

Generation of M. Put $L_n = A^m/I^{n+1}A^m$. Since $I^{n+1}M_n = 0$, φ_n factors through a surjection $\bar{\varphi}_n : L_n \to M_n$, compatible with the transition maps ($L_{n+1} \to L_n$ the projection), so we have a surjection of inverse systems with kernels $K'_n = \operatorname{Ker} \bar{\varphi}_n$. The maps $K'_{n+1} \to K'_n$ are surjective: let $a \in A^m$ with class in K'_n. Then $\varphi_{n+1}(a) \in \operatorname{Ker} \pi_n = I^{n+1}M_{n+1} = I^{n+1}\varphi_{n+1}(A^m) = \varphi_{n+1}(I^{n+1}A^m)$, so there is $b \in I^{n+1}A^m$ with $\varphi_{n+1}(a - b) = 0$; the class of $a - b$ lies in K'_{n+1} and maps to the class of a in L_n. By Lemma 3.1, $\varprojlim L_n \to \varprojlim M_n = M$ is surjective. Since A is complete, $\varprojlim A^m/I^{n+1}A^m = \widehat{A}^m = A^m$ (the map $A^m \to \varprojlim L_n$ is bijective), and the composite $A^m \to M$ is $a \mapsto (\varphi_n(a))_n = \sum a_i e_i$. So $M = \sum Ae_i$.

(c) Now A is Noetherian and M_0 finitely generated; by (b) M is finitely generated.

(1) For $x \in M$, $u_n(I^{n+1}x) \subseteq I^{n+1}M_n = 0$, so $I^{n+1}M \subseteq K_n$; and $K_{n+1} \subseteq K_n$ because $u_n = \pi_n u_{n+1}$. Hence $I^{n+1}M + K_{n+1} \subseteq K_n$. Conversely let $x \in K_n$. Then $u_{n+1}(x) \in \operatorname{Ker} \pi_n =$

$I^{n+1}M_{n+1} = u_{n+1}(I^{n+1}M)$ (as u_{n+1} is onto). So there is $z \in I^{n+1}M$ with $x - z \in K_{n+1}$, and $x \in I^{n+1}M + K_{n+1}$.

(2) Since $I^{n+1}M \subseteq K_n$, u_n induces $M/I^{n+1}M \to M_n$, compatible with transitions, hence $\lambda : \varprojlim M/I^{n+1}M \to \varprojlim M_n = M$. Let $\iota : M \to \varprojlim M/I^{n+1}M = \widehat{M}$ be the canonical map. Then $\lambda \circ \iota = \mathrm{id}_M$. As A is complete, $\widehat{A} = A$, so by Cor. 3.14 (M is finitely generated over the Noetherian ring A) $\iota : M = M \otimes_A A \to \widehat{M}$ is an isomorphism. Hence $\lambda = \iota^{-1}$ is an isomorphism.

(3) The inverse systems $(K_n/I^{n+1}M)$, $(M/I^{n+1}M)$, (M_n) with maps induced by the identity of M (so $K_{n+1}/I^{n+2}M \to K_n/I^{n+1}M$ is induced by inclusion) give an exact sequence $0 \to K_n/I^{n+1}M \to M/I^{n+1}M \to M_n \to 0$. By (1), the image of K_{n+1} in $K_n/I^{n+1}M$ is everything, so the transition maps of the first system are surjective. Lemma 3.1 gives an exact sequence

$$0 \to \varprojlim K_n/I^{n+1}M \to \varprojlim M/I^{n+1}M \xrightarrow{\lambda} \varprojlim M_n.$$

Since λ is injective by (2), $\varprojlim K_n/I^{n+1}M = 0$. For an inverse system with surjective transition maps, each projection from the limit is onto (lift step by step), so every $K_n/I^{n+1}M = 0$, i.e. $K_n = I^{n+1}M$. □

Exercise 3.12

> **Problem.** Let A be a Noetherian ring, I an ideal of A, and $\widehat{A}$ the I-adic completion of A. Let $a \in A$. Show that if a is not a zero divisor in A, then it is not a zero divisor in $\widehat{A}$. However, show, by giving an example, that A can be an integral domain without $\widehat{A}$ being one.

Solution. If a is a non-zero-divisor, multiplication $a : A \to A$ is injective. Since $\widehat{A}$ is flat over A (Thm. 3.15), tensoring with $\widehat{A}$ gives that multiplication by a on $A \otimes_A \widehat{A} = \widehat{A}$ is injective, i.e. a is not a zero divisor in $\widehat{A}$.

Example. Let k be a field of characteristic $\neq 2$ (e.g. $\mathbf{Q}$), $A = k[x,y]/(F)$ with $F = y^2 - x^2(x+1)$, and $I = (x,y)A$. F is irreducible in $k[x,y]$: otherwise, being monic of degree 2 in y over the UFD $k[x]$, it would have a root in $k(x)$ (Gauss), i.e. $(y/x)^2 = x + 1$ in $k(x)$, impossible since $x + 1$ has odd valuation at the prime $x + 1$. So (F) is prime and A is a domain. By Ex. 3.10, $\widehat{A} \simeq k[[u,v]]/(uv)$, which is not a domain: $u \notin (uv)$, $v \notin (uv)$, but $uv = 0$. □

Exercise 3.13

> **Problem.** Let A be a Noetherian ring, I an ideal of A, and M a finitely generated A-module. Suppose that A is complete for the I-adic topology. Show that every submodule of M is closed.

Solution. Let $N \subseteq M$ be a submodule; we must show $\bigcap_n (N + I^n M) = N$. Put $M' = M/N$, a finitely generated A-module; the intersection is the preimage of $\bigcap_n I^n M'$, so it suffices to show $\bigcap_n I^n M' = 0$.

Since A is complete, I lies in the Jacobson radical: for $\alpha \in I$ the series $\sum_{k \geq 0} (-\alpha)^k$ converges in A (partial sums are Cauchy) to an inverse of $1 + \alpha$ (the product with $1 + \alpha$ has partial

sums $1-(-\alpha)^{N+1}\to 1$, and A is separated). By Cor. 3.13, an element $z\in\bigcap_n I^nM'$ satisfies $(1+\alpha)z=0$ for some $\alpha\in I$; as $1+\alpha$ is a unit, $z=0$. Hence $\bigcap_n I^nM'=0$ and N is closed. □

Exercise 3.14

Problem. Let A be a Noetherian ring, I an ideal of A, and $\widehat{A}$ the formal I-adic completion of A. Show that the topology on $\widehat{A}$ induced by that on A is the $I\widehat{A}$-adic topology.

Solution. The topology on $\widehat{A}=\varprojlim A/I^n$ coming from that of A is the one defined (Section 1.3.1) by the filtration $\widehat{A}_n=\mathrm{Ker}(p_n:\widehat{A}\to A/I^n)$. We show $\widehat{A}_n=I^n\widehat{A}$; since $(I\widehat{A})^n=I^n\widehat{A}$, the two filtrations coincide.

Tensor the exact sequence $0\to I^n\to A\to A/I^n\to 0$ with $\widehat{A}$, which is flat (Thm. 3.15): the sequence $0\to I^n\otimes\widehat{A}\to\widehat{A}\to(A/I^n)\otimes_A\widehat{A}\to 0$ is exact, and the image of $I^n\otimes\widehat{A}$ in $\widehat{A}$ is $I^n\widehat{A}$. So $I^n\widehat{A}=\mathrm{Ker}(\widehat{A}\to(A/I^n)\otimes_A\widehat{A},\ \alpha\mapsto 1\otimes\alpha)$. Now A/I^n is a finitely generated A-module whose I-adic completion is itself (it is discrete: $I^k(A/I^n)=0$ for $k\geq n$), and by Cor. 3.14 the canonical map $(A/I^n)\otimes_A\widehat{A}\to A/I^n$, $b\otimes\alpha\mapsto b\,p_n(\alpha)$, is an isomorphism. The composite $\widehat{A}\to(A/I^n)\otimes\widehat{A}\to A/I^n$ sends $\alpha\mapsto p_n(\alpha)$. Therefore $I^n\widehat{A}=\mathrm{Ker}\,p_n=\widehat{A}_n$. □

Exercise 3.15

Problem. (*Mittag-Leffler condition*) Let $(A_n,\pi_n)_n$ be an inverse system of sets. For any pair $m>n$, let us denote the map $\pi_n\circ\pi_{n+1}\circ\cdots\circ\pi_{m-1}:A_m\to A_n$ by $\pi_{m,n}$. We say that the system $(A_n,\pi_n)_n$ satisfies the Mittag-Leffler condition if for every n, the descending sequence $(\pi_{m,n}(A_m))_{m>n}$ is stationary.

(a) Let $A'_n=\bigcap_{m>n}\pi_{m,n}(A_m)$. Show that $(A'_n,\pi_{n-1}|_{A'_n})_n$ is an inverse system and that the canonical map $\varprojlim_n A'_n\to\varprojlim_n A_n$ is bijective.

(b) Suppose that $(A_n,\pi_n)_n$ satisfies the Mittag-Leffler condition and that $A_n\neq\emptyset$ for all n. Show that $A'_n\neq\emptyset$, that $A'_{n+1}\to A'_n$ is surjective, and that $\varprojlim_n A'_n\neq\emptyset$. Deduce from this that $\varprojlim_n A_n\neq\emptyset$.

(c) Let $0\to(A_n)_n\to(B_n)_n\xrightarrow{\rho}(C_n)_n\to 0$ be an exact sequence of inverse systems of Abelian groups such that $(A_n)_n$ satisfies the Mittag-Leffler condition. Let $(c_n)_n\in\varprojlim_n C_n$ and $X_n=\rho_n^{-1}(c_n)$, where $\rho_n:B_n\to C_n$ is the n-th component of ρ. Show that $(X_n)_n$ is an inverse system of sets satisfying the Mittag-Leffler condition. Deduce from this that $\varprojlim_n B_n\to\varprojlim_n C_n$ is surjective. This generalizes Lemma 3.1.

Solution. (a) Let $x\in A'_n$. Then $\pi_{n-1}(x)\in\pi_{n,n-1}(A_n)$, and for $m>n$ we can write $x=\pi_{m,n}(y)$ with $y\in A_m$, so $\pi_{n-1}(x)=\pi_{m,n-1}(y)\in\pi_{m,n-1}(A_m)$. Thus $\pi_{n-1}(x)\in A'_{n-1}$, and (A'_n) is an inverse system. The canonical map $\varprojlim A'_n\to\varprojlim A_n$ is induced by the inclusions $A'_n\subseteq A_n$, hence injective. For surjectivity let $(a_n)\in\varprojlim A_n$. For all $m>n$, $a_n=\pi_{m,n}(a_m)\in\pi_{m,n}(A_m)$, so $a_n\in A'_n$ for every n, i.e. (a_n) comes from $\varprojlim A'_n$.

(b) By hypothesis, for each n there is $m(n)>n$ with $\pi_{m,n}(A_m)=A'_n$ for all $m\geq m(n)$ (the stationary value of the decreasing sequence is its intersection). As $A_m\neq\emptyset$, $A'_n\neq\emptyset$.

Surjectivity. Let $x\in A'_n$. Take $m\geq\max(m(n),m(n+1))$. Then $x\in\pi_{m,n}(A_m)$, say $x=\pi_{m,n}(y)$, $y\in A_m$. Put $z=\pi_{m,n+1}(y)$; then $z\in\pi_{m,n+1}(A_m)=A'_{n+1}$ and $\pi_n(z)=\pi_{m,n}(y)=x$.

Nonempty limit. Pick $a_0 \in A'_0$; given $a_n \in A'_n$, surjectivity of $A'_{n+1} \to A'_n$ gives $a_{n+1} \in A'_{n+1}$ with $\pi_n(a_{n+1}) = a_n$. Then $(a_n) \in \varprojlim A'_n \neq \emptyset$. By (a), $\varprojlim A_n \neq \emptyset$.

(c) Identify A_n with a subgroup of B_n (kernel of ρ_n). Write π_n also for the transition maps of B, C; ρ commutes with them and $\pi_n|_{A_{n+1}}$ is the transition map of A. If $b \in X_{n+1}$ then $\rho_n(\pi_n b) = \pi_n \rho_{n+1}(b) = \pi_n(c_{n+1}) = c_n$, so $\pi_n(X_{n+1}) \subseteq X_n$ and (X_n) is an inverse system of sets. Each X_n is nonempty (ρ_n is onto) and is a coset of A_n: $X_n = x_n + A_n$ for any $x_n \in X_n$.

Mittag-Leffler. For $m > n$, $\pi_{m,n}(X_m) = \pi_{m,n}(x_m) + \pi_{m,n}(A_m)$, a translate of the subgroup $\pi_{m,n}(A_m) \subseteq A_n$. Choose $m(n)$ such that $\pi_{m,n}(A_m) = A'_n$ for all $m \geq m(n)$. For $m \geq m(n)$ the sets $\pi_{m,n}(X_m)$ are cosets of the same subgroup A'_n of B_n, and they form a decreasing sequence. If a coset $x + H$ is contained in a coset $y + H$ of the same subgroup, then $x \in y + H$ and $x + H = y + H$. So the sequence is constant for $m \geq m(n)$. Hence (X_n) satisfies the Mittag-Leffler condition.

Surjectivity. The X_n are nonempty and satisfy Mittag-Leffler, so by (b) $\varprojlim X_n \neq \emptyset$. An element (b_n) of it is a compatible family with $\rho_n(b_n) = c_n$, i.e. an element of $\varprojlim B_n$ mapping to (c_n). Hence $\varprojlim B_n \to \varprojlim C_n$ is surjective. (If $A_{n+1} \to A_n$ is onto for all large n, then (A_n) satisfies Mittag-Leffler, so this contains Lemma 3.1.) $\square$